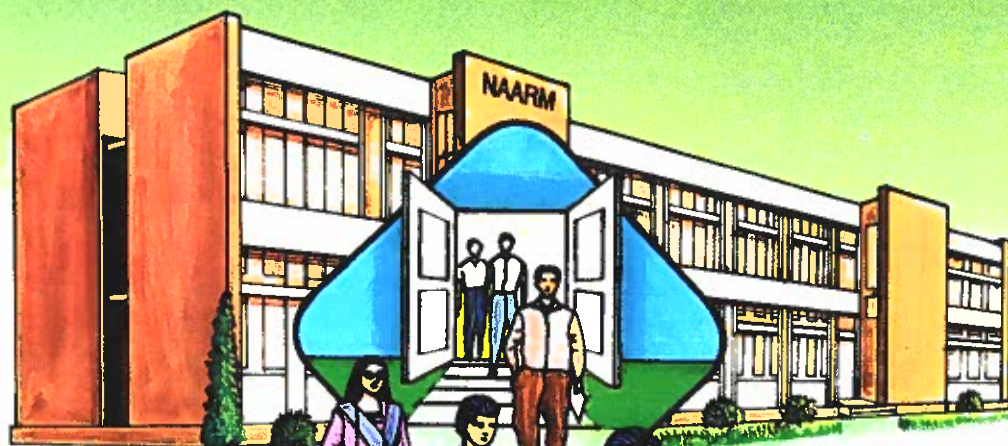


1998-99

वार्षिक प्रतिवेदन

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Building human resource for higher productivity



राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

राजेन्द्रनगर, हैदराबाद-500 030.

NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT

Rajendranagar, Hyderabad - 500 030

वार्षिक प्रतिवेदन
ANNUAL REPORT
1998-99



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Preface

The National Academy of Agricultural Research Management (NAARM) was established in 1976 for imparting management training to agricultural scientists of all levels with an aim to enhance their performance as agricultural research workers. The Academy is addressing issues of research management through different levels of training programmes, workshops, seminars, and policy-level retreat programmes. Since strengthening of NARS requires input from educational institutes, NAARM is focusing its attention on the problems of education management. The educational technology is very important aspect for the University teachers. Through regular training and sponsored programmes, the University teachers and the functionaries of different levels are being trained, the impact of which is quite visible. The impact study report has revealed that Universities like TNAU, TNVASU, and ANGRAU have been immensely benefited by our efforts.

In addition to Scientists and Teachers of Agricultural Universities and ICAR Institutes, the administrative and finance personnel are also trained in foundation courses and specially designed programmes for them.

The human resource development is the key element in enhancing continuous research performance in the Institutes. It is expected that in the next 20 years India will be able to assume leadership in production and productivity of several agricultural commodities. This projection is based on the trend of growth. However, in order to sustain this trend of growth, the leadership development in the scientific community is most crucial. The leadership with character, sense of responsibility and accountability, and mind set with attitude of service are the major emphasis in building future research workers. NAARM endeavours with belief that bringing change in the mind set is possible through training.

The faculty development efforts of our Institute is a continuous process and has given positive results. The present level of performance of faculty members of NAARM are commendable. The research studies taken up by them serve as resource material for the programmes conducted by them. Under AHRD Project, faculty members have been trained in India and abroad in different academic areas of research management. The participation of our faculty members in seminars and workshops conducted by the management institutes and agencies also helped in broadening the horizon of knowledge in their areas of interest.

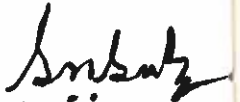
The guidance, support, and patronage we have received from Dr R.S.Paroda, Secretary, DARE, and Director General, ICAR, and Dr S.L.Mehta, Deputy Director General (Education), ICAR, and Dr A.K. Sinha, Assistant Director General (Education), ICAR, are of immense value for sustaining the activities of NAARM towards achieving excellence.

I express my gratitude for the dedicated service rendered by Dr P Manikandan, Dr N. Hanumantha Rao, Dr R. Kalpana Sastry, and Mr R.V.V.S. Prakasa Rao in compiling, designing, and bringing out the Annual Report in the present form. I acknowledge the services rendered

by Mr P. Namdev and Mr L. Venkateswarlu. The assistance received from Ms G. Aneeta in editing the theme paper is also acknowledged.

The co-operation and timely help from the faculty members and other officials are also duly acknowledged.

Hyderabad
October 1999


S.N. Saha
Acting Director

अप्रैल

डॉ. आर. एस. परोदा, महानिदेशक, भा. कृ. अनु. प. , नई दिल्ली की अध्यक्षता में अप्रैल 19, 1998 को भा. कृ. अनु. प.-इक्रीशैट (आई सी आर आई एस ए टी) संयुक्त नीति सलाहकार समिति की बैठक संपन्न हुई। भा. कृ. अनु. प. और इक्रीशैट के प्रतिनिधित्व करते हुए पच्चीस वरिष्ठ सदस्यों ने इसमें भाग लिया।

कृषि में कंप्यूटर प्रयोग पर दो सप्ताहीय पाठ्यक्रम का आयोजन अप्रैल 20 से मई 01, 1998 तक किया गया। इस पाठ्यक्रम में रा. कृ. वि. , भा. कृ. अनु. प. संस्थानों और अन्य संगठनों के कृषि से जुड़े हुए अट्वाइस वैज्ञानिक और तकनीकी अधिकारियों ने भाग लिया।

लोक उद्यम संस्थान, हैदराबाद के सहयोग से श्रमिक प्रबंध गतिकी पर अप्रैल 21-24, 1998 तक एक चार दिवसीय कार्यशाला का आयोजन किया गया। इस कार्यशाला में भा. कृ. अनु. प. संस्थानों के निदेशक, रा. कृ. वि. के प्रमुख निदेशक, फार्म प्रभारी अधिकारी और अन्य निर्णय लेने के पदों पर कार्यरत प्रशासकों ने भाग लिया जिनकी संख्या तीस थी। डॉ. आर. कल्पना शास्त्री और श्री. बी. पाण्डु रेड्डी इस कार्यशाला के समन्वयक थे।

मई

संकायाध्यक्षों के लिए, "कृषि शिक्षा का बेहतर प्रबंध कैसे करें ?" विषय पर मई 05-07, 1998 तक तीन दिवसीय कार्यशाला का आयोजन किया गया। इस कार्यशाला में विभिन्न कृषि विश्वविद्यालयों के संकायाध्यक्ष और भा. कृ. अनु. प. शिक्षण प्रभाग के सदस्यों ने भाग लिया जिनकी संख्या उनतीस थी। इस कार्यशाला के समन्वयक डॉ. एस. एन. साहा और डॉ. ए. गोपालम थे।

आचार्य एन. जी. रंगा कृषि विश्वविद्यालय के संकाय सदस्यों के लिए मई 19-31, 1998 तक दो दिवसीय शिक्षण प्रयोगिकी में संकाय विकास कार्यक्रम का आयोजन किया गया। इस कार्यक्रम में इकतीस सहायक एवं सह आचार्यों ने भाग लिया। इस कार्यक्रम के डॉ. ए. गोपालम, डॉ. जगन्नाथम चल्ला और डॉ. पी. कुमार समन्वयक थे।

इस अकादमी द्वारा मई 26-27, 1998 को राजभाषांतरण की समस्याएँ पर दो दिवसीय कार्यशाला का आयोजन किया गया। इस कार्यशाला में राजभाषा कार्यान्वयन से जुड़े बीस सहायक निदेशक (रा.भा), तकनीकी अधिकारी (हिन्दी) और अन्य अधिकारियों ने भाग लिया। इस कार्यशाला के समन्वयक डॉ. बी. एस. चन्देल थे।

इस अकादमी द्वारा आई बी एम ग्लोबल सर्विसेस इण्डिया लिमिटेड के सहयोग से राज्य कृषि विश्वविद्यालयों के सह आचार्य एवं सहायक आचार्यों के लिए शिक्षा शास्त्र पर दो कार्यशालाओं का आयोजन किया गया। पहली कार्यशाला का आयोजन मई 27-28, 1998 को और दूसरी कार्यशाला का आयोजन मई 29-30 को किया गया। इस कार्यशाला के समन्वयक डॉ. आर. चित्तिरैचेल्वन और डॉ. एस. के. नंदा थे।

जून

- * शिक्षा शास्त्र पर जून 08-09, 1998 को दो दिवसीय कार्यशाला का आयोजन किया गया। इस कार्यशाला में आचार्य एन. जी. रंगा कृषि विश्वविद्यालय और रा. कृ. अनु. प्र. अ. के 14 संकाय सदस्यों ने भाग लिया। डॉ. टेलर जानस्टन, मिचिगन स्टेट यूनिवर्सिटी, अमेरिका संयुक्त राज्य ने सुविधक की भूमिका निभाई। इस कार्यशाला के डॉ. एस. एन. साहा और डॉ. आर. चित्तिरैचेल्वन, समन्वयक थे।
- * जून 16-27, 1998 तक सामाजिक परिवर्तन के लिए सहभागी वीडियो पर दो सप्ताहीय पाठ्यक्रम का आयोजन किया गया। इस पाठ्यक्रम में भा. कृ. अनु. प. संस्थानों और रा. कृ. वि. के चार अधिकारियों ने भाग लिया। डॉ. आर. चित्तिरैचेल्वन और डॉ. वीई. सबारत्नम इस कार्यक्रम के समन्वयक थे।
- * आचार्य एन जी रंगा कृषि विश्वविद्यालय के संकाय सदस्यों के लिए शैक्षणिक प्रशासन में कार्यपालक विकास पर दस दिवसीय कार्यक्रम का आयोजन जून 22 से जुलाई 01, 1998 तक किया गया। इस कार्यक्रम में अठारह वरिष्ठ आचार्य और विभागाध्यक्षों ने भाग लिया। इस कार्यक्रम के डॉ. जगन्नाथम चल्ला और डॉ. एन. संध्या शेणाय समन्वयक थे।
- * जून 23 से जुलाई 04, 1998 तक कृषि में अनुसंधान और विस्तार के लिए शीघ्र, विश्रांत और सहभागी ग्रामोण मूल्यांकन तकनीक पर दो सप्ताहीय पाठ्यक्रम का आयोजन किया गया। इस कार्यक्रम में भा. कृ. अनु. प. संस्थानों और रा. कृ. वि. के वैज्ञानिक, कृषि विज्ञान केंद्र और कृषक प्रशिक्षण केंद्रों से प्रशिक्षकों ने भाग लिया जिनकी संख्या दस थी। इसके समन्वयक डॉ. वीई. सबारत्नम और डॉ. एन. संध्या शेणाय थे।
- * जून 29 से जुलाई 04, 1998 तक कृषि अनुसंधान में प्रबंध विकास कार्यक्रम का आयोजन किया गया। इस कार्यक्रम में भा. कृ. अनु. प. संस्थानों के तेरह प्रभागाध्यक्षों और भा. कृ. अनु. प. मुख्यालय के दो सहायक महानिदेशकों ने भाग लिया। इसके समन्वयक डॉ. एस. एन. साहा और डॉ. जगन्नाथम चल्ला थे।

जुलाई

- * जुलाई 14-25, 1998 तक कृषि अनुसंधान के लिए प्रबंध सूचना पद्धति पर एक पाठ्यक्रम का आयोजन किया गया। इस कार्यक्रम में भा. कृ. अनु. प. संस्थानों, रा. कृ. वि. और अन्य अनुसंधान एवं विकास संस्थानों में कृषि अनुसंधान से जुड़े हुए सत्रह वैज्ञानिक और तकनीकी अधिकारियों ने भाग लिया। इस कार्यक्रम के डॉ. टी. बालगुरु और डॉ. पी. मणिकण्डन समन्वयक थे।
- * जुलाई 15-18, 1998 तक राष्ट्रीय कृषि अनुसंधान पद्धति में स्वाट (एस डब्ल्यू ओ टी) विश्लेषण का प्रयोग पर चार दिवसीय कार्यक्रम का आयोजन किया गया। इस पाठ्यक्रम में भारतीय राष्ट्रीय कृषि अनुसंधान पद्धति से संबंधित पच्चीस वैज्ञानिक कार्मिकों ने भाग लिया। इस कार्यक्रम के डॉ. बी. एस. चंदेल और डॉ. के. वेंकटेश्वर्लु समन्वयक थे।

कार्यशाला में
र जानस्टन,
ई. एस. एन.

का आयोजन
का डॉ. आर.

लक विकास
ठारह वरिष्ठ
य समन्वयक

भागो ग्रामीण
संस्थानों और
की संख्या दस

इस कार्यक्रम
देशकों ने भाग

का आयोजन
स्थानों में कृषि
बालगुरु और

षण का प्रयोग
द्विती से संबंधित
समन्वयक थे।

जुलाई 20-24, 1998 तक भा. कृ. अनु. प. -आई. सी. आई. एम. लिमिटेड द्वारा संयुक्त रूप से कानफिगरेशन ग्राफ सन सोलरिस सर्वर पर पाँच दिवसीय कार्यक्रम का आयोजन किया गया। इस कार्यक्रम के द्वारा ग्यारह सहभागियों को प्रशिक्षित किया गया।

तमिलनाडु कृषि विश्वविद्यालय के संकाय सदस्यों के लिए दि. 21 जुलाई से 3 अगस्त, 1998 तक शिक्षा प्रौद्योगिकी में दो सप्ताह का संकाय विकास कार्यक्रम आयोजित किया गया। इस कार्यक्रम में पच्चीस सहायक आचार्य एवं सह आचार्यों ने भाग लिया। डॉ. ए. गोपालम, डॉ. एस. एन. साहा, तथा डॉ. आर. कल्पना शास्त्री ने इस कार्यक्रम का समन्वयन किया।

पुस्तकालय एवं सूचना केन्द्र में कंप्यूटर प्रयोग पर दस दिवसीय प्रशिक्षण कार्यक्रम दि. 27 जुलाई से 6 अगस्त 1998 तक आयोजित किया गया। भा. कृ. अनु. प., राज्य कृषि विश्वविद्यालय तथा अन्य संगठनों के चौदह प्रशिक्षण अधिकारी, प्रलेख अधिकारी एवं सूचना से जुड़े व्यवसायी वर्ग ने इस कार्यक्रम में भाग लिया। डॉ. के. विद्यासागर राव इस कार्यक्रम के समन्वयक रहे।

भा. कृ. अनु. प. के कृषि अनुसंधान प्रशिक्षण एवं विस्तार संगठनों में प्रशासनिक सक्षमता में सुधार पर दि. 29 जुलाई से 04 अगस्त, 1998 तक एक सप्ताह का पाठ्यक्रम आयोजित किया गया। इस कार्यक्रम में कुल अड़सठ प्रशासनिक एवं लेखा से जुड़े कर्मचारियों ने भाग लिया। डॉ. एस. एन. साहा एवं श्री. बी. पाण्डु रेड्डी ने इस कार्यक्रम का समन्वयन किया।

अगस्त

विज्ञान एवं प्रौद्योगिकी विभाग की परियोजना के अन्तर्गत कृषि में व्यवसायी महिलाओं की, चौथी परियोजना में संबंधित सलाहकार समिति की बैठक 4 अगस्त, 1998 को आयोजित हुई। इस बैठक में सात सदस्यों ने भाग लिया।

कृषि व्यवसाय प्रबंध पर 17-22 अगस्त, 1998 तक एक प्रशिक्षण कार्यक्रम आयोजित किया गया। भा. कृ. अनु. प. तथा राज्य कृषि विश्वविद्यालयों के वैज्ञानिक, कृषि व्यवसायी कंपनी तथा सरकारी निगम के कार्यकारी वर्ग, कृषि व्यवसाय से जुड़े व्यवसायी तथा परामर्श फर्मों से जुड़े कार्यकारी कुल 17 लोगों ने इस कार्यक्रम में भाग लिया। इसका समन्वयन डॉ. के. पी. सी. राव ने किया।

दि. 17-29 अगस्त 1998 तक डेटा विश्लेषण के लिए सांख्यिकीय साफ्टवेयर पर कार्यक्रम आयोजित किया गया। इस कार्यक्रम में भा. कृ. अनु. प., राज्य कृषि विश्वविद्यालय तथा अन्य संगठनों के सात वैज्ञानिक तथा तकनीकी अधिकारियों ने भाग लिया। डॉ. के. विद्यासागर राव ने इस कार्यक्रम का समन्वयन किया।

अकादमी में दि. 31 अगस्त से 11 सितंबर, 1998 तक शैक्षणिक वीडियो पर दो सप्ताह का प्रशिक्षण कार्यक्रम आयोजित किया गया। इस कार्यक्रम में भा. कृ. अनु. प., राज्य कृषि विश्वविद्यालय तथा अन्य सरकारी विभाग जिनको

शैक्षणिक वीडियो निर्माण में रुचि हो, ऐसे सोलह वैज्ञानिक तथा तकनीकी अधिकारियों ने भाग लिया। इस कार्यक्रम का समन्वयन श्री. के. आर. प्रभाकर ने किया।

सितंबर

* कृषि अनुसंधान एवं विकास समस्याओं की पहचान एवं निदान हेतु सहभागिता पद्धति पर सितंबर 01-05, 1998 तक पाँच दिवसीय प्रशिक्षण कार्यक्रम एवं कार्यशाला का आयोजन किया गया। राष्ट्रीय कृषि अनुसंधान केंद्र के विभिन्न संस्थानों से कुल सात वैज्ञानिक और अधिकारियों ने इस कार्यक्रम में भाग लिया। डॉ. एन. संध्या शेणाय और डॉ. वीई. सबारत्नम इस कार्यक्रम के समन्वयक रहे।

* सितंबर 07-18, 1998 तक प्रबंध सूचना पद्धति पर प्रशिक्षण पाठ्यक्रम आयोजित किया गया। राष्ट्रीय कृषि अनुसंधान केंद्रों के सूचना केंद्रों से जुड़े वरिष्ठ अधिकारी, ए आर आई एस विभाग के प्रभारी अधिकारी, राष्ट्रीय कृषि प्रौद्योगिकी परियोजना के सूचना आधारित परियोजना प्रबंधकों ने इस पाठ्यक्रम में भाग लिया। इस पाठ्यक्रम का समन्वयन डॉ. अजीत पी. मारु ने किया तथा सत्रह सहभागियों ने इसमें भाग लिया।

* सितंबर 07-18, 1998 तक मानव संसाधन प्रबंध पर दो सप्ताह का पाठ्यक्रम आयोजित किया गया। इस कार्यक्रम में विभिन्न संगठनों के ग्यारह वैज्ञानिक एवं अधिकारियों ने भाग लिया। डॉ. एम. एम. अनवर एवं डॉ. आर. कल्पना शास्त्री इस पाठ्यक्रम के समन्वयन रहे।

* प्रशिक्षकों के प्रशिक्षण पर सितंबर 15-26, 1998 तक दो सप्ताह का पाठ्यक्रम आयोजित किया गया। इस पाठ्यक्रम के माध्यम से चार प्रशिक्षकों को प्रशिक्षित किया गया। डॉ. पी. मणिकण्डन एवं डॉ. टी. बालगुरु ने इस पाठ्यक्रम का समन्वयन किया।

अक्तूबर

* अक्तूबर 06-17, 1998 के दौरान कृषि में कंप्यूटर प्रयोग पर दो सप्ताह का प्रशिक्षण कार्यक्रम आयोजित किया गया। इस कार्यक्रम में राज्य कृषि विश्वविद्यालय, भा. कृ. अनु. प. संस्थान, तथा अन्यसंगठनों के अठारह वैज्ञानिक एवं तकनीकी अधिकारियों ने भाग लिया। डॉ. के. विद्यासागर राव एवं डॉ. के. एम. रेड्डी इस कार्यक्रम के समन्वयक रहे।

* कृषि शिक्षा प्रबंध पर अक्तूबर 29-31, 1998 तक तीन दिवसीय कार्यशाला आयोजित की गई। राज्य कृषि विश्वविद्यालय के 30 संकायाध्यक्षों ने इस कार्यशाला में भाग लिया। डॉ. एस. एन. साहा एवं डॉ. ए. गोपालम ने इस कार्यशाला का समन्वयन किया।

* वैयक्तिक एवं संगठनात्मक प्रभावत्मकता के लिए कृषि वैज्ञानिक विकास पर 21-31 अक्तूबर, 1998 के दौरान दस दिवसीय प्रशिक्षण कार्यक्रम आयोजित किया गया। इस कार्यक्रम में भा. कृ. अनु. प. संस्थान, राज्य कृषि

1. इस कार्यक्रम

विश्वविद्यालय, राज्य सरकारी विभाग, गैरसरकारी अधिकारी कुल तेरह वरिष्ठ स्तर के सहभागियों ने भाग लिया। यह कार्यक्रम डॉ. एस. शण्मुगम् एवं डॉ. जगन्नाथम चल्ला द्वारा समन्वित किया गया।

नवम्बर

दिसंबर 01-05,
अनुसंधान केंद्र
संध्या शेणाय

अकादमी के द्वारा टी. आई एफ ए सी, नई दिल्ली के सहयोग से प्रौद्योगिकी पूर्वानुमान एवं मूल्य निर्धारण पर 09-13 नवम्बर 1998 तक एक पाँच दिवसीय प्रशिक्षण कार्यक्रम आयोजित किया गया। इस कार्यक्रम का समन्वयन डॉ. डी. रामाराव एवं डॉ. एस. के. नंदा द्वारा किया गया जिसमें विभिन्न अनुसंधान एवं विकास संस्थानों के 14 सहभागियों ने भाग लिया।

दिसंबर

राष्ट्रीय कृषि
राष्ट्रीय कृषि
पाठ्यक्रम का

आचार्य एन. जी. रंगा कृषि विश्वविद्यालय के लिए दि. 1 से 11 दिसंबर 1998 तक शैक्षणिक प्रशासन एवं प्रबंध पर दस दिवसीय कार्यकारी विकास कार्यक्रम आयोजित किया गया। कुल बीस सहायक आचार्य व सह आचार्यों ने इस कार्यक्रम में भाग लिया, जिसके समन्वयक डॉ. ए. गोपालम एवं डॉ. आर. कल्पना शास्त्री रहे।

किया गया। इस
एवं डॉ. आर.

कृषि अनुसंधान उत्पादकता प्रबंध पर दस दिवसीय पाठ्यक्रम दिसंबर 01-11, 1998 तक आयोजित किया गया। इसके समन्वयक डॉ. पी. कुमार और डॉ. वीई. सबारत्नम थे। इसमें भा. कृ. अनु. प. संस्थानों और रा. कृ. वि. के आठ वैज्ञानिक और संकाय सदस्यों ने भाग लिया।

किया गया। इस
इस पाठ्यक्रम

गृहविज्ञान शिक्षा पर दो दिवसीय मस्तिष्क मंथन (ब्रेनस्टार्मिंग) कार्यशाला दिसंबर 02-03, 1998 तक आयोजित की गई। विभिन्न विश्वविद्यालयों के कुल सैंतीस सदस्यों जिनमें कुलपति, संकायाध्यक्ष और वरिष्ठ संकाय सदस्यों ने इसमें भाग लिया। डॉ. करोल मीक्स, संकायाध्यक्ष, कॉलेज ऑफ फैमली एण्ड कन्ज्यूमर साइन्सेज, आई ओ डब्ल्यू ए स्टेट यूनिवर्सिटी ने इस कार्यक्रम में संसाधन व्यक्ति के रूप में भाग लिया। डॉ. डी. रामाराव और डॉ. एन. संध्या शेणाय इस पाठ्यक्रम के समन्वयक थे।

आयोजित किया
ह वैज्ञानिक एवं
समन्वयक रहे।

निरन्तर कृषि विकास के लिए उत्पादन पद्धति उपागम पर पाँच दिवसीय कार्यशाला दिसंबर, 07-11, 1998 तक आयोजित की गई। एन ए आर एस के विभिन्न संगठनों के सात अनुसंधानकर्ता और समाज वैज्ञानिकों ने इसमें भाग लिया। डॉ. टी. बालगुरु और डॉ. पी. मणिकण्डन ने कार्यक्रम का समन्वयन किया।

गई। राज्य कृषि
इस कार्यशाला

उन्नत डाटा विश्लेषण हेतु सांख्यिकीय साफ्टवेयर पर दस दिवसीय कार्यक्रम दिसंबर 21-31, 1998 तक आयोजित किया गया। भा. कृ. अनु. प., रा. कृ. वि. और अन्य आर एण्ड डी संस्थानों के तेरह वैज्ञानिक और तकनीकी अधिकारियों ने इसमें भाग लिया। इसका समन्वयन डॉ. के. विद्यासागर राव ने किया।

1998 के दौरान
न, राज्य कृषि

राजभाषा कार्यान्वयन में अनुभवजन्य प्रयोग पर दो दिवसीय कार्यशाला दिसंबर 28-29, 1998 तक आयोजित की गई। भा. कृ. अनु. प. संस्थानों के हिन्दी अनुभाग के बयालीस अधिकारियों ने इसमें भाग लिया।

जनवरी,

- * तमिलनाडु पशुचिकित्सा और पशु विज्ञान विश्वविद्यालय, चेन्नई के संकाय सदस्यों हेतु तीन सप्ताह का शिक्षा प्रौद्योगिकी पर संकाय विकास कार्यक्रम जनवरी 04-22, 1999 तक आयोजित किया गया। इसमें चौबोस सह आचार्यों ने भाग लिया। इसका समन्वयन डॉ. ए. गोपालम, डॉ. पी. कुमार और डॉ. एस. के. नंदा ने किया।
- * कृषि अनुसंधान प्रबंध पर तीन दिवसीय कार्यपालक विकास कार्यक्रम जनवरी 28-30, 1999 तक आयोजित किया गया। इसमें चौदह निदेशकों ने भाग लिया। डॉ. एस. एन. साहा और डॉ. जगन्नाथम चल्ला ने इसका समन्वयन किया।

फरवरी

- * तमिलनाडु पशुचिकित्सा और पशु विज्ञान विश्वविद्यालय, चेन्नई के संकाय सदस्यों हेतु फरवरी 01-18, 1999 तीन सप्ताह का शिक्षा प्रौद्योगिकी पर संकाय विकास कार्यक्रम आयोजित किया गया। इसमें सत्ताईस सह आचार्यों ने भाग लिया। डॉ. ए. गोपालम, डॉ. के. पी. सी. राव और डॉ. बी. एस. चन्देल ने इस कार्यक्रम का समन्वयन किया।
- * ग्लोबल लर्निंग सेंटर और इंटरनेट आधारित लर्निंग कोर्सवेयर के विकास हेतु दो सप्ताह का पाठ्यक्रम फरवरी 08-20, 1999 तक आयोजित किया गया। इसमें विभिन्न संगठनों के सात अध्यापक और ए आर आई एस से जुड़े दो तकनीकी अधिकारियों ने भाग लिया। इसका समन्वयन डॉ. अजीत पी. मारू ने किया।
- * कृषि अनुसंधान प्रबंध के सुदूर प्रशिक्षण हेतु अनुसंधान परियोजना के परामर्श समूह की प्रथम बैठक 18 फरवरी, 1999 को आयोजित की गई। समूह के अध्यक्ष डॉ. एस. एल. मेहता, उप-महानिदेशक (शिक्षा), भा.कृ.अनु.प. नई दिल्ली ने बैठक की अध्यक्षता की। डॉ. अब्दुल खान, कुलपति, आई जी एन ओ यू, नई दिल्ली; डॉ. जे. सी. कत्याल, निदेशक, नार्म; डॉ. पाल पेराल्ट, भारत में ईन्ट्रार प्रतिनिधि; डॉ. एस. एन. साहा, संयुक्त निदेशक, नार्म; डॉ. ए. पद्मराज, अनुसंधान निदेशक, ए एन जी आर ए यू, हैदराबाद और डॉ. जगन्नाथम चल्ला, परियोजना के प्रधान अनुसंधानकर्ता ने बैठक में भाग लिया।

मार्च

- * सूचना केंद्रों के प्रबंध पर दो सप्ताह का पाठ्यक्रम मार्च 08-20, 1999 तक आयोजित किया गया। ए आर आई एस से सीधे जुड़े विभिन्न संगठनों के ग्यारह सहभागियों ने इसमें भाग लिया। डॉ. अजीत पी. मारू इसके समन्वयक थे।
- * कृषि शिक्षा प्रबंध पर तीन दिवसीय कार्यशाला मार्च 19-21, 1999 तक आयोजित की गई। विभिन्न रा. कृ. वि. के छब्बीस संकायाध्यक्षों ने इसमें सक्रिय रूप से भाग लिया। इसका समन्वयन डॉ. ए. गोपालम और डॉ. एस. एन. साहा ने किया।

Activities at a Glance

April

* ICAR - ICRISAT Joint Policy Advisory Committee Meeting was held on April 19, 1998, under the Chairmanship of Dr R.S. Paroda, Director General, ICAR, New Delhi. Twenty five senior members, representing ICAR and ICRISAT, attended the meeting.

* A two-week course on Computer Applications in Agriculture was organized from April 20 to May 1, 1998. Twenty eight scientists and technical officers of SAUs, ICAR institutes, and other organizations related to agriculture participated in the course. Dr K.M. Reddy and Dr K. Vidyasagar Rao were the co-ordinators of the course.

* A four-day workshop on Dynamics of Labour Management was organized from April 21 to 24, 1998, in collaboration with the Institute of Public Enterprise, Hyderabad. Directors of ICAR institutes, Directors of Research from SAUs, Officers-in-charge of farm, and other administrators in decision-making positions, numbering 30, participated in the workshop. Dr H. Kalpana Sastry and Mr B. Pandu Reddy co-ordinated the workshop.

May

* A three-day Deans workshop on 'How to Manage Agricultural Education Better ?' was organized from May 5 to 7, 1998. Deans of various Agricultural Universities and members from Education Division of ICAR, numbering 29, participated in the workshop. Dr S.N. Saha and Dr A. Gopalram co-ordinated the workshop.

* A two-week Faculty Development Programme in Educational Technology for the faculty members of Acharya N.G. Ranga Agricultural University was organized from May 19 to 31, 1998. Thirty one Assistant and Associate Professors attended the programme. Dr A. Gopalram, Dr Jagannadham Challa, and Dr P. Kumar co-ordinated the programme.

* A two-day workshop on "Rajabhashaantaran kee Samasyaayen" (Problems related with shifting to Hindi as an Official Language) was organized at the Academy on May 26 and 27, 1998. Twenty technical officers and other officers, working in the area of implementation of official language, participated in the workshop. Dr B.S. Chandel was the co-ordinator of the workshop.

* Two workshops on Pedagogy were organized by the Academy in collaboration with IBM Global Services India Ltd. for the Associate and Assistant Professors of State Agricultural Universities. The first workshop was organized on May 27 and 28, and the second one was organized from May 29 to 30, 1998. Dr R. Chithiraichelvan and Dr S.K. Nanda co-ordinated these workshops.

June

- * A two-day workshop on Pedagogy was organized on June 8 and 9, 1998. Faculty members from Acharya N.G. Ranga Agricultural University and NAARM, numbering 14, participated in the workshop. Dr Taylor Johnston from Michigan State University, USA, acted as the facilitator. Dr S.N. Saha and Dr R.Chithiraichelvan were the co-ordinators of this workshop.
- * A two-week course on Participatory Video for Social Change was organized from June 16 to 27, 1998. Four officials from ICAR institutes and SAUs participated in the course. Dr R. Chithiraichelvan and Dr VE. Sabarathnam co-ordinated the programme.
- * A ten-day Executive Development Programme in Educational Administration and Management, for the faculty members of Acharya N.G. Ranga Agricultural University, was organized from June 22 to July 1, 1998. Eighteen senior professors and Heads of the Departments participated in this programme. Dr Jagannadham Challa and Dr N. Sandhya Shenoy were the co-ordinators of this programme.
- * A two-week course on Rapid, Relaxed, and Participatory Rural Appraisal Techniques for Research and Extension in Agriculture was organized from June 23 to July 4, 1998. Scientists from ICAR institutes and SAUs, trainers from Krishi Vigyan Kendras and Farmers Training Centres, numbering 10, participated in this programme, which was co-ordinated by Dr VE. Sabarathnam and Dr N. Sandhya Shenoy.
- * Management Development Programme in Agricultural Research was organized from June 29 to July 4, 1998. Thirteen Heads of the Divisions from ICAR institutes and two ADGs from ICAR Headquarters participated in this programme, co-ordinated by Dr S.N. Saha and Dr Jagannadham Challa.

July

- * A course on Management Information System for Agricultural Research was organized from July 14 to 25, 1998. Seventeen scientists and technical officers from ICAR institutes, SAUs, and other R & D institutions, involved in agricultural research, participated in the programme. Dr T. Balaguru and Dr P. Manikandan were the co-ordinators of this programme.
- * A four-day programme on Application of SWOT Analysis in National Agricultural Research System was organized from July 15 to 18, 1998. Twenty five scientific personnel belonging to Indian NARS participated in the programme. Dr B.S. Chandel and Dr K. Venkateswarlu co-ordinated the programme.
- * A five-day training programme on Configuration of Sun Solaris Server was jointly organized by ICAR and ICIM Ltd., at the Academy, from July 20 to 24, 1998. Eleven participants were trained through this programme.

A two-week Faculty Development Programme in Educational Technology, for the faculty members of Tamil Nadu Agricultural University, was organized from July 21 to August 3, 1998. Twenty five Assistant Professors and Associate Professors participated in the programme. Dr A. Gopalam, Dr S.N. Saha, and Dr R. Kalpana Sastry co-ordinated the programme.

A ten-day training programme on Computer Application in Libraries and Information Centres was organized from July 27 to August 6, 1998. Fourteen librarians, documentation officers, and information professionals from ICAR institutes, SAUs, and other organizations participated in the programme. Dr K. Vidyasagar Rao was the co-ordinator of this programme.

A one-week course on Improving Administrative Efficiency in Agricultural Research Training and Extension Organizations of ICAR was organized from July 29 to August 4, 1998. A total of 68 administrative and accounts personnel attended this programme. Dr S.N. Saha and Mr B. Pandu Reddy co-ordinated the programme.

August

The Fourth PAC (Project Advisory Committee) meeting of DST Project on Professional Women in Agriculture was held on August 4, 1998. The meeting was attended by seven members.

A training programme on Agri-business Management was organized from August 17 to 22, 1998. Executives of agri-business companies and government corporations, scientists of ICAR institutes and SAUs, professionals interested in agri-business, and personnel from consultancy firms, numbering 17, participated in the programme, co-ordinated by Dr K.P.C. Rao.

A programme on Statistical Software for Data Analysis was organized from August 17 to 29, 1998. Seven scientists and technical officers from ICAR institutes, SAUs, and other organizations participated in the programme. Dr K. Vidyasagar Rao co-ordinated the programme.

A two-week training programme on Educational Video Production was organized at the Academy from August 31 to September 11, 1998. Sixteen scientists and technical officers from ICAR institutes, SAUs, and other government departments, interested in production of educational video materials, attended the programme. Mr K.R. Prabhakar was the co-ordinator of the programme.

September

A five-day training-cum-workshop on Participatory Methods for Identification and Diagnosis of Agricultural Research and Development Problems was organized from September 1 to 5, 1998. Scientists and officials from various institutions of NARS, numbering seven, participated in the workshop. Dr N. Sandhya Shenoy and Dr VE. Sabarathnam co-ordinated the programme.

- * A training course on Management of Information Centres was organized from September 7 to 18, 1998. The participants for this course comprised senior officers managing information centres in NARS, officers-in-charge of ARIS Cell, and information-related project managers of National Agricultural Technology Project. The course, co-ordinated by Dr Ajit P. Maru, was attended by 17 participants.
- * A two-week course on Management of Human Resources was organized from September 7 to 18, 1998. Eleven scientists and officers from various organizations participated in the programme. Dr M.M. Anwer and Dr R. Kalpana Sastry co-ordinated the course.
- * A two-week course on Training of Trainers was organized from September 15 to 26, 1998. Four trainers were trained through this course. Dr P. Manikandan and Dr T. Balaguru co-ordinated the programme.

October

- * A two-week training programme on Computer Applications in Agriculture was organized from October 6 to 17, 1998. Eighteen scientists and technical officers from SAUs, ICAR institutes, and other organizations related to agriculture participated in the programme. Dr K. Vidyasagar Rao and Dr K.M. Reddy were the co-ordinators of this programme.
- * A three-day workshop on Management of Agricultural Education was organized from October 29 to 31, 1998. Thirty Deans from various State Agricultural Universities participated in the workshop. Dr S.N. Saha and Dr A. Gopalam co-ordinated the workshop.
- * A ten-day training programme on Agricultural Scientist Development for Personal and Organizational Effectiveness was organized from October 21 to 31, 1998. Thirteen senior-level participants from ICAR institutes, SAUs, State Departments, and NGOs participated in this programme, which was co-ordinated by Dr S. Shanmugam and Dr Jagannadham Challa.

November

- * A five-day training programme on Technological Forecasting and Assessment was organized by the Academy in collaboration with TIFAC, New Delhi, from November 9 to 13, 1998. The programme, co-ordinated by Dr D. Rama Rao and Dr S.K. Nanda, was attended by 14 participants from various R & D institutes.

December

- * A ten-day Executive Development Programme in Educational Administration and Management was organized for Acharya N.G. Ranga Agricultural University from December 1 to 11, 1998. Twenty Assistant Professors and Associate Professors participated in the programme, which was co-ordinated by Dr A. Gopalam and Dr R. Kalpana Sastry.

A ten-day course on Management of Agricultural Research Productivity was organized from December 1 to 11, 1998. The course, co-ordinated by Dr P. Kumar and Dr VE. Sabarathnam, was attended by eight scientists and faculty members from various ICAR institutes and SAUs.

A two-day Brainstorming Workshop on Home Science Education was organized on December 2 and 3, 1998. Vice Chancellors, Deans, and senior faculty members from various universities, numbering 37, attended the workshop. Dr Carol Meeks, Dean, College of Family and Consumers Sciences, Iowa State University, USA, participated in the programme as a resource person. Dr D. Rama Rao and Dr N. Sandhya Shenoy co-ordinated the programme.

A five-day course on Production Systems Approach for Sustainable Agricultural Development was organized from December 7 to 11, 1998. Seven researchers and social scientists, belonging to different organizations of NARS, participated in the programme. Dr T. Balaguru and Dr P. Manikandan co-ordinated the programme.

A ten-day programme on Statistical Software for Advanced Data Analysis was organized from December 21 to 31, 1998. Thirteen scientists and technical officers from ICAR institutes, SAUs, and other R & D institutions were benefited by this course, co-ordinated by Dr K. Vidyasagar Rao.

A two-day workshop on Rajbhasha Karyanvayan Mein Anubhavjanya Prayog was organized on December 28 and 29, 1998. Forty two officials from Hindi Cells of various ICAR institutes participated in the workshop.

January 1999

A three-week Faculty Development Programme in Educational Technology was organized from January 4 to 22, 1999, for the faculty members of the Tamil Nadu Veterinary and Animal Sciences University, Chennai. Twenty four Assistant Professors participated in the programme, co-ordinated by Dr A. Gopalam, Dr P. Kumar, and S.K. Nanda.

A three-day Executive Development Programme in Agricultural Research Management was organized from January 28 to 30, 1999. Fourteen Directors attended the programme. Dr N. Saha and Dr Jagannadham Challa co-ordinated the programme.

February 1999

A three-week Faculty Development Programme in Educational Technology was organized from February 1 to 18, 1999, for the faculty members of the Tamil Nadu Veterinary and Animal Sciences University, Chennai. Twenty seven Assistant Professors participated in the programme. Dr A. Gopalam, Dr K.P.C. Rao, and Dr B.S. Chandel co-ordinated the programme.

A two-week course on Development of Global Learning Centres and Internet-based

Learning Courseware was organized from February 8 to 20, 1999. Seven teachers and two technical officers associated with ARIS, belonging to various organizations, were benefited by participation in this course, co-ordinated by Dr Ajit P. Maru.

* The first meeting of the Project Advisory Group of the Research Project on Distance Training for Agricultural Research Management was held on February 18, 1999. The Chairman of the group Dr S.L. Mehta, Dy. Director General, ICAR, New Delhi, presided over the meeting. Dr J.C. Katyal, Director, NAARM; Dr S.N. Saha, Joint Director, NAARM; Dr Paul Perrault, ISNAR representative in India; Dr Abdul Khan, Vice Chancellor, IGNOU, New Delhi; Dr A. Padmaraju, Director of Research, ANGRAU, Hyderabad; and Dr Jagannadham Challa, Principal Investigator of the project, attended the meeting.

March 1999

* A two-week course on Management of Information Centres was organized from March 8 to 20, 1999. Eleven participants, involved directly in ARIS, from various organizations, participated in the course. Dr Ajit P. Maru co-ordinated the programme.

* A three-day workshop on Management of Agricultural Education was organized from March 19 to 20, 1999. The programme had the active participation of 26 Deans from various SAUs. This workshop was co-ordinated by Dr A. Gopalam and Dr S.N. Saha.

The National Academy of Agricultural Research Management (NAARM) was established in 1976, at Rajendranagar, Hyderabad, Andhra Pradesh, to fulfill the important need for an institution of management in agricultural research and education. NAARM is one of the five National Institutes of the Indian Council of Agricultural Research (ICAR), New Delhi, the apex body of the country for promoting agricultural research, education, and extension education.

The Academy is located in the premises of Acharya N.G. Ranga Agricultural University in Rajendranagar. The Academy, spread over 50 hectares area, has enviable sylvan environs. Over the years, the Academy has been able to build up excellent infrastructure and facilities necessary for imparting quality training. The faculty members of the Academy are extensively trained across a range of areas covering agricultural research and education management, both in India and abroad. With the excellent facilities available for training and with the expertise and experience available with the faculty members, the Academy has been able to build its strength for taking up varied activities to fulfil its mandate.

Since its inception, the Academy has taken up the onerous responsibility of developing the vast human resource engaged in agricultural research, education, and technology transfer functions in the country. The Academy organizes training programmes, Executive Development Programmes, workshops, and seminars in the areas related to agricultural research, technology transfer, and education management for the scientists at different levels in the National Agricultural Research System (NARS). Investigations on management problems associated with agricultural research and education form another major activity of the Academy. The results from such research effort form the basis for training as well as for organizing workshops and seminars. In addition, the Academy offers consultancy services in the area of its mandated activities.

The Academy's programmes are designed in such a way as to generate a sense of fraternity and inculcate scientific culture among the agricultural scientists in the country. With commitment and sincerity as the watch-words, the Academy has been able to elevate its image and credibility in the national and international arena. In addition to the programmes organized for the Indian scientists, the Academy also sets up tailor-made international programmes in agricultural research and education management. The Academy is looked upon as a model institute by the developing countries and it is fast developing into a viable Regional Training Centre in Asia to train agricultural scientists and academicians in agricultural research and education management.

Mandate

- ◆ To organize and conduct training programmes in agricultural research management for the scientists at various levels
- ◆ To build up high quality resource material in agricultural research management based on actual field experience
- ◆ To undertake systematic review and study of management problems of agricultural research institutes, programmes, and systems
- ◆ To plan, organize, and conduct workshops and seminars in research management and educational technology
- ◆ To organize, liaise, and co-ordinate programmes of international co-operation in the field of agricultural research management
- ◆ To assist the State Agricultural Universities (SAUs) in developing Regional Centres of Management to cater to the needs of various Universities

Objectives

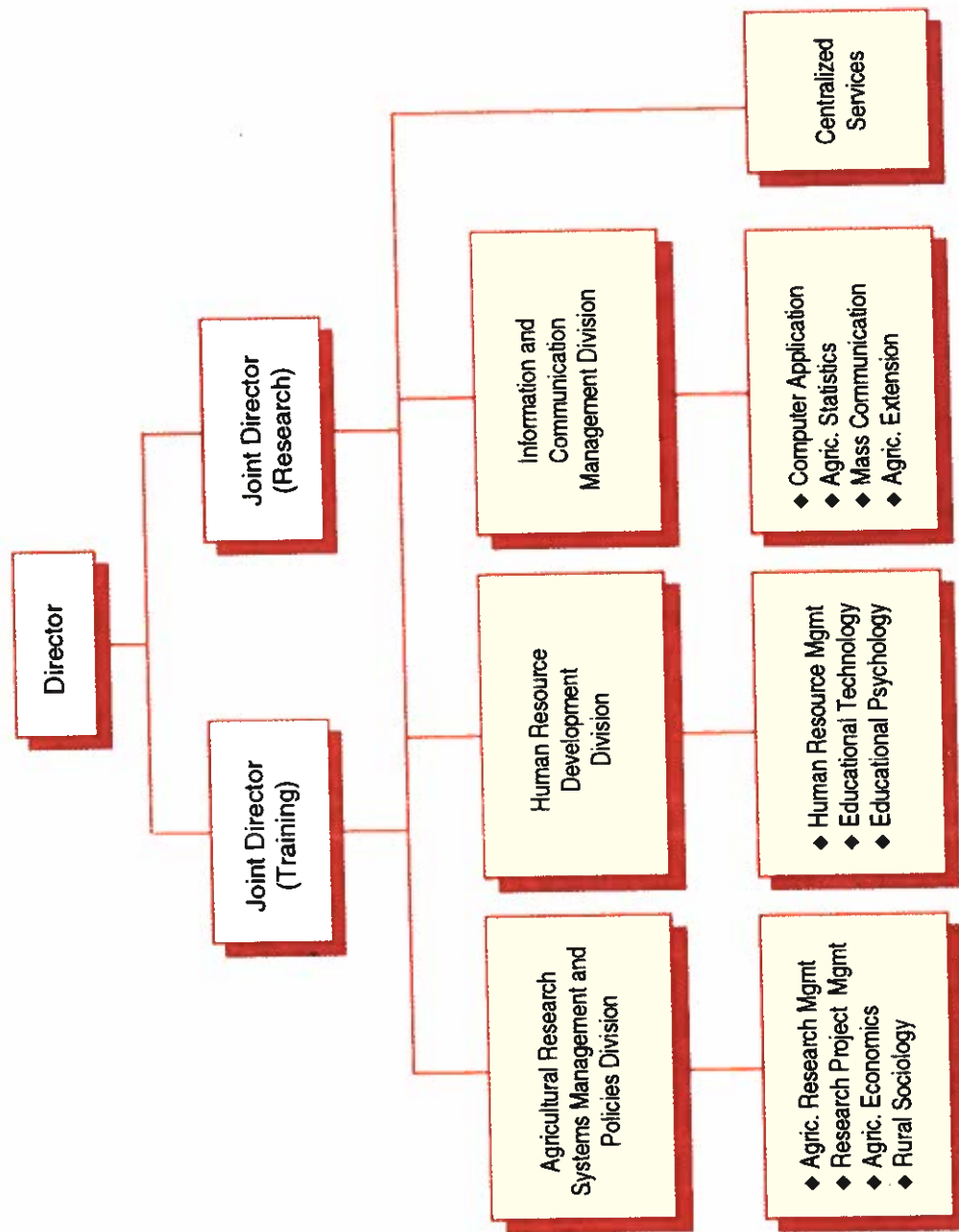
- ◆ To organize foundation courses for the new entrants of scientists to the Agricultural Research Service of ICAR
- ◆ To design and organize training programmes, workshops, and seminars, and to develop high quality material in agricultural research and education management for the senior and middle-level scientists of NARS
- ◆ To play a major role in different processes of human resource development of NARS in the context of agricultural research and education
- ◆ To undertake policy-level studies on prioritization of research and resource allocation, and to enhance research productivity of the constituents of NARS and their programmes
- ◆ To evolve suitable management techniques for effective planning, scheduling, organizing, monitoring, and evaluation of agricultural research and development projects in NARS
- ◆ To study the management issues of various programmes and patterns of technology transfer, develop effective systems and strategies for effective communication of agricultural technologies

- To undertake development programmes for evolving more effective and efficient approaches, methods, and models in higher agricultural education for use in SAUs and Deemed Universities of ICAR
- To promote and execute academic programmes of higher learning in the specialized areas of agricultural research and education management
- To develop expertise in information technology to meet the application software needs of NARS
- To undertake systematic review and to develop appropriate systems for efficient and effective administrative and financial management in NARS
- To function as a repository of ideas and information in the form of databases in the field of agricultural research and education management, and to act as a clearing house for information dissemination through national and international networks in these areas
- To offer consultancy services in the field of agricultural research and education management, and agricultural information and communication technology

Organizational Structure

The Academy is headed by the Director, who is guided and supported by the Institute Management Committee (IMC). The IMC reviews and promotes programmes of the Academy, and approves investments in new areas of research and education, training programmes, workshops, and seminars. The Research Advisory Committee (RAC) and Staff Research Council (SRC) provide broad guidelines and assist in developing and implementing specific research programmes and projects. The Director is assisted by two Joint Directors (currently one Joint Director is in position) and three Division Heads in the execution of various programmes. The Academy is organized into three functional Divisions, viz. Agricultural Research Systems Management and Policies, Human Resource Development, and Information and Communication Management. A multi-disciplinary team of scientists, who also constitute the core faculty of the Academy, works in different Divisions. Each faculty member is an agricultural scientist, with specialization in one of the disciplines of agriculture, who has also received advanced and specialized training in the broad area of management. As needed and appropriate, the Academy borrows the services of outside resource persons, specialized in specific areas, for its multifarious training programmes. The functioning of the Academy is supported by a number of centralized services such as Administration and Accounts, Hostel Services, Health Services, and Press and Public Relations Cell. The current organizational structure of NAARM is presented in the following figure.

Organizational Structure of NAARM



Leadership

Leadership could be defined in terms of the totality of functions performed by the executives as individuals and as a group, or it may be viewed as the problem of techniques, authority and position, authority relationship, leadership styles, communication, interpretation and enforcement, socio-psychological environment, organizational and personal goal conflicts, etc. The functions of the individuals and organizations revolve around decisions taken by the executive leadership which has influence and impact on the system. Therefore, the specific action in a specific situation cannot be determined without the knowledge of the many factors that are involved in organizational management. Appropriate action in one situation may not be appropriate in another situation. This very concept calls for a clear understanding of the environment and situation, and accordingly adopt strategies for enhancing the performance of the organization. One cannot expect any sort of perfect solution to the leadership and authority problems. It must be considered that complex interpersonal relationships of socio-psychological factors do not remain constant over time. Leadership may, in practice, mean almost infinite number of possible combinations of variables. It involves skills that cannot be neatly categorized into a logical pattern, for purposes of education and training.

Quality Leadership

It is difficult to explain and understand the inherent qualities possessed by highly effective leaders in different fields. They seem to violate theoretical precepts of leadership for the same reason that several aspects of natural phenomena do not fit the theories. In other words, a true leader does not need a definite set of rules, but a method of analyzing the organizational situation in which he must act. Leadership can be understood as an inherent or acquired capacity to frame plans that will succeed, and the faculty to pursue others to carry them out in the face of difficulties or bottlenecks. The first part of this concept is that one should have the capacity to form plans, programmes, projects, etc. that have a probability of success in diverse environmental situations. It implies that the plan will be realistic. For doing so, a leader should have full information about the size and quality of his resources including men, material, and functions. He should also understand the environment in which the plans have to be implemented. The second part deals with the implementation. No plan can ever be implemented exactly as originally conceived and framed. It has to be implemented by those who are working for their leader. A leader should have the ability to persuade others to implement the plans despite difficulties, discouragement, and setbacks. It is this capability which amounts to greater probability in the effectiveness of a leader. There are two fundamental issues related to the capability of a leader. These are i) Who can be a good leader? and ii) What does leadership process involve?

Paper prepared by Dr S. N. Saha, Acting Director, NAARM

Unit 1

Mass Communication
Agric. Extension

Educational Psychology

Agric. Economics
Rural Sociology

Executives cannot become leaders unless they possess the faculty of good thoughts, words, and deeds that are in harmony. The thoughts should not be from the source of his lust, anger, attachment, greed, egotism, or jealousy. Thus, the leadership process has been summed up into four stages, viz. to be, to do, to see, and to tell. Decision-making is the most important element of leadership, and this capability is essential in institutional management. The elementary ingredients in this process are:

- 1) being absolutely clear about the goal that has to be achieved; collecting and evaluating all related information about the men and material resources available for the task; and
- 2) the environment in which the task has to be carried out; opposition and obstacles that are likely to be met and the time that is available for completing the task, crystallizing the options that emerge from the evaluated information and examining the advantages; selecting an option and formulating the plan or the project for implementation; ability to make use of the creativity of people working with him.

The quality leadership deals with the consistency and flexibility in functional leadership. However, one can begin to see them in a new light. They can be interpreted as helping or hindering three areas: achieving task, building and maintaining them, and developing individuals. Some qualities are especially important because they apply to all three areas.

Quality Attributes

Enthusiasm is an excellent example. Those who have the gift of enthusiasm almost always will spark it off in other people. It produces greater commitment to task, creates team spirit, and enthuses the individual. Some qualities are latent, and they can be made to surface out. Moral, courage, and humility are required in certain situations. But it is important to be as specific as possible in defining when they are needed. Humility may seem as an odd word because it implies to many people as a cringing self-abasement quite at odds with self-confidence, even egoism. The qualities like initiative, integrity, humour, tact, etc. are most important for personality. Aristotle taught us long ago that a virtue rests somewhere between two extremes. If you have any quality in excess, or without the moderating influence of balancing qualities it can become a liability. Certainly, too much humility or rather humility of the counterfeit sort is fatal to leadership; for it robs self-confidence that one should have. A leader should own the responsibility of mistake, not blame others.

In the leader, we find distinct styles of leadership: one is authoritarian and the other is democratic. The authoritarian leader makes all the decisions with little, if any, subordinate participation. The democratic leader actively solicits suggestions from subordinates, frequently acts on their advice, and gives them a range of discretion in performing their activities with decentralization in decision making. The assumption is frequently made that the authoritarian leader is primarily concerned with the accomplishment of tasks, and that the democratic leader has a particular concern for the people. The distinction between manager and leaders, thus, becomes clear. Managers are inclined towards tasks whereas leaders concern is simultaneous

development of people. The next century must look for leaders rather than managers.

The shift in concept and practice is taking place very fast to match with the technological innovations in the areas of information generation, transmission, and utilization. The challenges in front of the agricultural scientists and other workers are enormous, and they need to adapt these changes at appropriately faster rate to make the system pro-active, and to position the organization for future requirements. For organizations to survive in future, they must not only be convinced that being lead by the best and brightest helps the organization but also that without the best and the brightest at the helm, the chance for survival is nil. Those organizations will be the survivors in future that are smarter, faster, and more innovative, and they will have the people at the top who think that way. But thinking alone is not enough. It will take know-how and courage to implement and to carry on their convictions, i.e. their plans. A visionary planner and a task-master for implementation of the plan will be the requirements for the top leadership. It will take more than knowing the pre-requisites to perform the effective management and leadership. The Chief Executive Officers (CEOs) must identify and eliminate the obstacles to effective management and leadership. The unwillingness of the top leadership to subjugate their personal lives to the operational demands needs to be changed through rigorous developmental training. There is a conflict between the new generation which resents the idea that the best bureaucrats, as opposed to the best performers, are more likely to get ahead. It is a common evidence that the rewards go to the best politician who is not necessarily the best executive. People may be encouraged to be creative if the top leadership wears the same hat and the same coat as every body else. The hypocrisy is the most dangerous syndrome of weak and inefficient leadership. Professional performance, with professional zeal, should be the key for the leadership.

Leadership Empowers People

The empowerment of people in the organization in terms of releasing the intrinsic enthusiasm and enjoyment in the work will be the major task of the leadership. The next century will look forward to the leadership performance of the executives of organizations for their survival and relevant contribution to the society.

Leadership Success

The major cause of poor performance by the leaders may be traced to dishonesty, arrogance, focus on personal advancement, ego, rigidity, my way is the only way syndrome, failure to develop people to the fullest, and failure to build trust amongst people. For effective functioning of leadership, task needs, individual needs, and group needs are the most important elements on which appropriate focus should be made. In practical terms, it is to achieve task, its mission, objectives, or goals. For doing so, one has to build his team as a cohesive group and develop every individual in the team to give his very best. Consequently, he has to harmonize and integrate the needs related to the accomplishment of the task with those of the group he leads and the individuals in the group. Setting standards by personal example, maintaining self-discipline, building team spirit, training the group, knowing individuals personally, etc. are some of the determinants for leadership performance. The following steps may be useful for improving

the effectiveness.

- Being crystal clear about the task that is to be achieved
- Planning to achieve the task by using the available resources and people
- Controlling by managing the work
- Supporting by encouragement, recognition, and training people
- Evaluating to learn lessons so that performance can be improved

Leaders identify and articulate the central concern of people and awaken them at a particular point of time for performance towards achieving the quality output. Managers do the things right, leaders do the right things. These two phrases are able to explain the conceptual difference between a leader and a manager. A leader constantly makes all efforts to train himself to be a proper leader for others. The followers will always look up at the leaders with emotional attachment. Thus, the following four attributes of an effective leader are most important.

- ❑ Professional knowledge and competence
- ❑ Ability to make up mind to take decisions and accept total responsibility for the decision
- ❑ Absolute firmness and impartiality in implementing the decisions
- ❑ Moral and physical courage

Better leaders are those who adopt the behaviour, attitude, and attributes that are normally the characteristics of women. A successful leader will always be collaborative to followers and directs upon communication. Good and respected leaders are those who

- ❖ Create opportunities for others to succeed
- ❖ Empower those who are close to the work to take their own decision
- ❖ Encourage their staff and support them when necessary
- ❖ Achieve the performance for which the team exists
- ❖ Develop people and teams
- ❖ Set an example by their own behaviour
- ❖ Build relationships on trust

It may be understood that any employee of the organization can lead, provided he has the critical knowledge base that can alter the academic directions.

Executive Decisions

Decisions are made by the different levels of executives. But the question remains to be addressed as to how to convey these decisions to the people. The superior executives may help the subordinate executives in making decisions. But it is always appropriate that privilege of communicating the decisions should be with the subordinate executives. The executives have

the working capital by virtue of their position and authority. But the leadership qualities need to be developed in them. Basically, the executive may not be a good leader. However, the inherent capabilities can be developed with consistent development process. The authority positions of the executive must be enforced. Also, at the same time, the leadership behaviour of the executive will have to be matched with the task and the situation. The individual accountability for quality output will be the major concern of a leader of any organization.

In the 21st century, the organizational survival depends upon the leadership performance of executives and managers.

Training Programmes

Programme	Date	Activity
Foundation Courses		
Foundation Course for Agricultural Research Service		The Foundation Course for Agricultural Research Service (FOCARS), which is a basic induction training for the scientists newly recruited to Agricultural Research Service, is organized in three phases. Phase 1 : Agricultural research systems and management processes - concepts and principles; Phase 2 : Technology generation and transfer process - field experience training; and Phase 3 : Agricultural research systems and management processes - practice and applications. The foundation course aims at exposing the entry-level scientists of ARS, to the various aspects of agricultural research project management, in addition to developing adequate skills in them for managing their research, by inculcating inter-disciplinary team spirit. It also prepares them to respond to the future demands of agricultural research at the national as well as at the international level.
FOCARS 62	07-01-98 05-05-98	
FOCARS 63	05-06-98 30-09-98	
FOCARS 64	21-07-98 13-11-98	
FOCARS 65	07-10-98 03-02-99	
FOCARS 66	06-01-99 05-05-99	During the year under report, five foundation courses were organized, which included the later part of FOCARS 62 and the earlier part of FOCARS 66. Dr T. Balaguru and Dr S. Shanmugam; Dr Jagannadham Challa and Dr P. Manikandan; Dr M.M. Anwer and Dr D. Rama Rao; Dr VE. Sabarathnam and Dr K. Vidyasagar Rao; and Dr K.P.C. Rao and Dr R. Chithiraichelvan were the co-ordinators of these various foundation courses.
Senior-level Programmes		
Computer Applications in Agriculture	20-04-98 01-05-98 06-10-98 17-10-98	The course aimed at exposing the participants to the facilities offered by MS-Office and other packages, and developing skills in them for application of these packages in their work. Two programmes were conducted during the year under report.
		A total of 46 scientists and technical officers from ICAR institutes and SAUs were benefited by participation in these two programmes. Dr K.M. Reddy and Dr K. Vidyasagar Rao co-ordinated the first programme and the second programme was co-ordinated by Dr K. Vidyasagar Rao and Dr K.M. Reddy.
Participatory Video for Social Change	16-06-98 27-06-98	The course on Participatory Video for Social Change aimed at exposing the participants to the participatory approaches in communication and to provide hands-on-experience in producing participatory video for social change. Various topics related to participatory approaches, team effort and small group behaviour, and participatory video production were covered during the course.

Four officials from ICAR institutes and SAUs participated in the course, co-ordinated by Dr R. Chithiraichelvan and Dr VE. Sabarathnam.

The application of rapid, relaxed, and participatory rural appraisal techniques for the purpose of agricultural research and extension is gaining momentum now-a-days. In order to impart the required skill in the participants to use various appraisal techniques in the actual field situation and to enable them to link these techniques for project preparation in research and extension, a course on Rapid, Relaxed, and Participatory Rural Appraisal Techniques for Research and Extension in Agriculture was organized. The programme enabled the participants to learn various appraisal techniques through field visit, exercises, and conceptual lectures.

Scientists from ICAR institutes and SAUs, trainers from Krishi Vigyan Kendras and Farmers Training Centres, numbering 10, participated in this programme, which was co-ordinated by Dr VE. Sabarathnam and Dr N. Sandhya Shenoy.

The Management Development Programme in Agricultural Research was organized with the main objective to develop management competencies at the middle level in the organizational set-up. Four themes, viz. research management, human resource management, administrative and financial management, and information and communication technology were covered through lectures, interactive discussion, case study, and group presentation.

Thirteen Heads of Divisions from ICAR institutes and two ADGs from ICAR Headquarters participated in the programme, co-ordinated by Dr S.N. Saha and Dr Jagannadham Challa.

Agricultural research institutions are faced with the challenge of efficient and effective management of resources, which are becoming scarce in view of increased institutional activities. Relevant and timely information is an essential pre-requisite for effective planning, programming, monitoring, and evaluation of research. It has, therefore, become necessary to develop an organized Management Information System (MIS) which can generate information for good decision-making. Keeping this in mind, a programme on Management Information System for Agricultural Research was organized. The programme acquainted

		the participants with the concept of MIS for agricultural research and assisted them to develop a computed-based MIS towards better research management in agriculture.
		Seventeen scientists and technical officers from ICAR institutes, SAUs, and other R & D institutions involved in agricultural research participated in the programme. Dr T. Balaguru and Dr P. Manikandan were the co-ordinators of this programme.
Application of SWOT Analysis in National Agricultural Research System	15-07-98 18-07-98	The programme acquainted the participants with the concept of SWOT - its application to individuals and households; and private and government organizations, with special reference to the National Agricultural Research System.
		Twenty five scientific personnel belonging to Indian NARS participated in the programme. Dr B.S. Chandel and Dr K. Venkateswarlu co-ordinated the programme.
Configuration of Sun Solaris Server	20-07-98 24-07-98	The programme was jointly organized by ICAR and ICIM Ltd., at the Academy.
		Eleven participants were trained by ICIM through this programme. The training was conducted on the Unix Operating System and on the operation of Sun Solaris Server.
Computer Applications in Libraries and Information Centres	27-07-98 07-08-98	The programme on Computer Applications in Libraries and Information Centres was organized to familiarize the managers of information in the NARS to the various packages of information management and to develop expertise in them for application in their work situation.
		Fourteen librarians, documentation officers, and information professionals from ICAR institutes, SAUs, and other organizations participated in the programme. Dr K. Vidyasagar Rao was the co-ordinator of this programme.
Improving Administrative Efficiency in Agricultural Research, Training, and Extension Organizations of ICAR	29-07-98 04-08-98	A programme on Improving Administrative Efficiency in Agricultural Research, Training, and Extension Organizations of ICAR was organized to improve the skills and efficiency of the administrative and accounts



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personnel in ICAR in discharging their functions and in providing constructive support to the Directors in meeting the organizational goals and needs.

A total of 68 administrative and accounts personnel attended this programme. Dr S.N. Saha and Mr B. Pandu Reddy co-ordinated the programme.

Programme 17-08-98
on Agri- 22-08-98
business
Manage-
ment

With the new economic policies initiated in 1991 and with the World Trade Agreement signed in 1994, agri-business management has gained a lot of importance. Keeping in view the need to assess the prospects of agri-business enterprises in the country, a programme on Agri-business Management was organized to apprise the participants of the problems, constraints, and challenges faced by agri-business sector and also of the policy initiatives needed to promote agri-business activities.

Executives of agri-business companies and government corporations, scientists of ICAR institutes and SAUs, professionals interested in agri-business, and personnel from consultancy firms, numbering 17, participated in this programme, co-ordinated by Dr K.P.C. Rao.

Statistical 17-08-98
Software 29-08-98
for Data
Analysis

The course on Statistical Software for Data Analysis was organized to familiarize the scientists to work on statistical software packages for data analysis and to provide adequate training on presentation of data and interpretation of results. The various topics that were dealt with included MSTAT C, MICROSTAT, MS Excel, Graphic packages, and GIST.

Seven scientists and technical officers from ICAR institutes, SAUs, and other organizations attended the programme. Dr K. Vidyasagar Rao was the co-ordinator of this programme.

Educational 31-08-98
Video Pro- 11-09-98
duction

A programme on Educational Video Production was organized with a view to developing skills in the participants in the production of educational video programmes and to familiarize them with the creative aspects of making educational videos. The participants were given sufficient theoretical background and hands-on-experience for effective planning, script writing, shooting, and editing in order to make creative educational video programmes.

Sixteen scientists and technical officers from ICAR institutes, SAUs, and other government departments, interested in the

Participatory Methods for Identification and Diagnosis of Agricultural Research and Development Problems

01-09-98
05-09-98

production of educational video materials, attended the programme. Mr K.R. Prabhakar was the co-ordinator of the programme.

A training-cum-workshop on Participatory Methods for Identification and Diagnosis of Agricultural Research and Development Problems was organized to expose the participants to the philosophy of PRA and to impart skills for developing location-specific action plans.

Faculty members and scientists from SAUs, ICAR institutes, and Krishi Vigyan Kendras, and officials from State Departments, numbering seven, participated in the programme. Dr N. Sandhya Shenoy and Dr VE. Sabarathnam were the co-ordinators of this programme.

Management of Human Resources

07-09-98
18-09-98

Management of human resources has become a very crucial element in organizations. This calls for increased attention to management of human resources for improving the organizational performance. To cater to this specific requirement, a course on



Management of Human Resources was organized. The programme aimed at sensitizing the managers to the issues of human behaviour in the organization by exposing them to the theories and practices of human behaviour. The various topics covered and the various instruments administered during the programme helped the participants to develop a better understanding of the individual, group, and organizational behaviour.

Eleven scientists and officers from various organizations participated in the programme. Dr M.M. Anwer and Dr R. Kalpana Sastry co-ordinated the programme.

Management of Information Centres

07-09-98
18-09-98
08-03-99
20-03-99

Information Centres in organizations are systems developed to meet the needs of every individual organization and can participate and function smoothly in global information networks. With the rapid change in information and communication technology, there is a need to train people managing information centres

and resources in their organizations. With a view to providing an opportunity for such personnel to exchange their experience and skills, two programmes were organized. These programmes covered a broad area of important issues in managing information centres ranging from hardware to skills enhancement in information technology and information management in relation to organizations. These programmes enabled the participants to develop their own webpages and websites, and publish them on NAARM Webserver.

A total of 28 participants were benefited by these two programmes, co-ordinated by Dr Ajit P. Maru.

Training of Trainers 15-09-98
26-09-98

The course was organized with the objectives of exposing the trainers to the concept and use of adult learning theory and experiential learning cycle, and developing adequate skills in them to prepare and conduct training programmes and individual sessions effectively. Various topics related to planning, organizing, and conducting training programmes as well as individual training sessions were covered through theoretical orientation, simulation exercises, individual presentations, and group practice sessions and video feedback. Most of the sessions were covered following co-training methodology which helped the participants to see and learn the effectiveness of group training and group teaching.

Four trainers were trained through this course. Dr P. Manikandan and Dr T. Balaguru co-ordinated the course.

Agricultural Scientist Development for Personal and Organizational Effectiveness 21-10-98
31-10-98

Scientists play a vital role as a facilitator to achieve organizational effectiveness. There is, therefore, a need to familiarize the scientists with the concepts and skills for enhancing personal and organizational effectiveness. In order to provide an opportunity for the scientists to develop the needed knowledge and skills, a course on Agricultural Scientist Development for Personal and Organizational Effectiveness was organized.

Thirteen senior-level scientists from ICAR institutes, SAUs, State Departments, and NGOs participated in this programme, which was co-ordinated by Dr S. Shanmugam and Dr Jagannadham Challa.

Technological Forecasting and Assessment	09-11-98 13-11-98	Perspective planning and future studies are essential for the growth and development of agricultural research organizations. Orientation to future studies requires specialized skills. In order to develop skilled personnel for application of forecasting and assessment in agriculture, a specialized course on Technological Forecasting and Assessment (TF/TA) was organized in collaboration with TIFAC, New Delhi. Conceptual lectures, case studies, group discussions, and exercises were used in the programme to orient the participants to the concept of TF/TA and to develop skills in them for the use of the tools and techniques of TF/TA. The programme, co-ordinated by Dr D. Rama Rao and Dr S.K. Nanda, was attended by 14 scientists from various R & D institutes.
Management of Agricultural Research Productivity	01-12-98 11-12-98	In order to facilitate the agricultural scientists to understand and practice various skills essential for improving research performance and productivity, a course on Management of Agricultural Research Productivity was organized. The various topics covered in the course included Research management skills, Research evaluation and assessment, Generating appropriate technologies, and Computer applications in research and research management. The course, co-ordinated by Dr P. Kumar and Dr VE. Sabarathnam, was attended by eight scientists and faculty members from ICAR institutes and State Agricultural Universities.
Production Systems Approach for Sustainable Agricultural Development	07-12-98 11-12-98	With the shift from commodity research towards production systems research and with the increased emphasis being laid on sustainability of production systems, it has become necessary to develop a clear understanding of production systems approach and the processes involved in its implementation, among all those concerned with sustainable agricultural development. Keeping this in view, a course on Production Systems Approach for Sustainable Agricultural Development was organized. The course aimed to familiarize the participants with the basic concept and process of production systems approach, the state-of-art of its implementation, and the measures needed for effective implementation. Dr T. Balaguru and Dr P. Manikandan co-ordinated this programme. Seven researchers and social scientists, belonging to different organizations of NARS, participated in the programme.

Statistical 21-12-98
Software for 31-12-98
Advanced
Data Analy-
sis

The advent of user-friendly software packages has made the analysis of complex and voluminous data simple. The advanced packages facilitate the scientists perform complicated analysis by themselves. With a view to familiarize the scientists on the statistical softwares for advanced data analysis and to develop necessary skills in them to handle the statistical packages for analysis of data and presentation of results more effectively, a Course on Statistical Software for Advanced Data Analysis was organized. Various aspects covered in the programme included Experimental design, stability analysis, and non-orthogonal data analysis; Multivariate analysis; Forecasting techniques; Statistical models; and various other advanced software packages for statistical analysis.

Thirteen scientists and technical officers of ICAR institutes, SAUs, and other R & D institutions were benefited by this course, co-ordinated by Dr K. Vidyasagar Rao.

Executive 28-01-99
Develop- 30-01-99
ment
Programme
in Agricul-
tural
Research
Manage-
ment

The Executive Development Programme in Agricultural Research Management aimed at examining the challenges of general institute management, information management, and organizational effectiveness. The programme covered various themes, viz. Institute management, Administration management, Financial management, Labour management, and other service rules pertaining to management competencies. Concept awareness lectures, interactive discussions, case study analysis, and group discussions were used to cover the different themes.

Fourteen Directors participated in this programme, co-ordinated by Dr S.N. Saha and Dr Jagannadham Challa.

Develop- 08-02-99
ment of 20-02-99
Global
Learning
Centres
and
Internet-
based
Learning
Courseware

With the development and growth of the Internet, spectacular improvements have been brought about in education over the last few years. The need for any time, anywhere, and just-in-time learning has become a reality with the development and use of information technology. With a view to develop a core group of teachers and trainers adept at using new computer - mediated information technology in agricultural education and extension, a course on Development of Global Learning Centres and Internet-Based Learning Courseware was organized.

Seven teachers and two technical officers associated with ARIS, belonging to various organizations, attended the programme, which was co-ordinated by Dr Ajit P. Maru.

Sponsored Programmes

Executive Development Programmes in Educational Administration and Management

22-06-98
01-07-98
01-12-98
11-12-98

Two Executive Development Programmes in Educational Administration and Management were organized, exclusively for the faculty members of Acharya N.G.Ranga Agricultural University, Hyderabad, with the main objective to develop management capacity in the faculty members in the area of educational administration and management. The programme covered various topics like Research, financial, and human resource management, Labour employment and conflict resolution, Information technology and management, Team building and team working, Infrastructure development and maintenance, Decentralization and delegation of powers, etc. to provide a broader perspective on important areas of management.

A total of 38 faculty members at various levels in the university were benefited by participation in these two programmes. Dr Jagannadham Challa and Dr N. Sandhya Shenoy co-ordinated the first programme, whereas the second programme was co-ordinated by Dr A. Gopalam and Dr R. Kalpana Sastry.

Faculty Development Programme in Educational Technology

19-05-98
02-06-98
21-07-98
03-08-98
04-01-99
22-01-99
01-02-99
18-02-99

Under the teacher orientation series, four Faculty Development Programmes in Educational Technology were organized for the faculty members of Acharya N.G. Ranga Agricultural University, Hyderabad (Programme 1), Tamil Nadu Agricultural University, Coimbatore (Programme 2), and Tamil Nadu Veterinary and Animal Sciences University, Chennai (Programmes 3 and 4). These tailor-made programmes emphasized on equipping the teachers to be more effective, and on developing the teachers to be competent resource persons for the institution. These programmes facilitated the faculty members to see and learn the art and skill of effective teaching, which requires a judicious combination of knowledge with various other related skills.

In all, 107 faculty members belonging to these universities were trained through these programmes. Dr A. Gopalam, Dr Jagannadham Challa, and Dr P. Kumar co-ordinated the first programme. The second programme was co-ordinated by Dr A. Gopalam, Dr S.N. Saha, and Dr R. Kalpana Sastry. Dr A. Gopalam, Dr P. Kumar, and Dr S.K. Nanda were the co-ordinators of the third programme. Dr A. Gopalam, Dr K.P.C. Rao, and Dr B.S. Chandel co-ordinated the fourth one.



Dynamics of Labour Management 21-04-98
24-04-98

A four-day workshop on Dynamics of Labour Management was organized in collaboration with the Institute of Public Enterprise (IPE), Hyderabad, to discuss the dynamics of agricultural labour management in the context of globalization of agriculture. Issues related to government regulations, employment and management of labour, personal and ecological safety measures, role of employers and employees, and various other aspects related to labour management were discussed during the workshop.



Directors of ICAR institutes, Directors of Research from State Agricultural Universities, Officers-in-Charge of farm, and other administrators in decision-making positions, numbering 30, participated in the workshop. Dr R. Kalpana Sastry and Mr B. Pandu Reddy from NAARM, and Dr P. Jyothi from IPE co-ordinated the programme.

How to Manage Agricultural Education Better? 05-05-98
07-05-98

A three-day Deans workshop on "How to Manage Agricultural Education Better?" was organized at the Academy. The workshop provided a forum for discussing various issues that need to be taken into account for effective management of Agricultural Education.

Deans from various SAUs and members from the Education Division of ICAR, numbering 29, participated in the workshop. Dr S.N.Saha and Dr A. Gopalam co-ordinated the workshop.

Rajabhasha antaran Kee Samasyaayen 26-05-98
27-05-98

A two-day workshop on "Rajabhashaantaran Kee Samasyaayen" (Problems related with shifting to Hindi as Official Language) was organized. The objective of the workshop was to discuss the problems associated with shifting to Hindi as Official Language and also to make the participants aware of all the provisions of the Official Language Act. The workshop also aimed at discussing the problems related to transformation in the context of globalization and privatization.

Twenty technical officers and other officers, working in the area of implementation of official language, participated in the workshop. Dr B.S. Chandel was the co-ordinator of the workshop.

Workshop on Pedagogy 27-05-98
28-05-98

29-05-98
30-05-98

08-06-98
09-06-98

These workshops covered the essentials of pedagogy with a view to help the teachers organize their experience in a framework. It also helped them extend their repertoire of teaching techniques and tools.



The first two workshops were organized in collaboration with IBM Global Services India Ltd. Dr R. Chithiraichelvan and Dr S.K. Nanda were the co-ordinators of these workshops. Dr Taylor Johnston from Michigan State University, USA, acted as the facilitator for the third workshop, which was co-ordinated by Dr S.N.Saha and Dr R. Chithiraichelvan. In all, 34 faculty members from various SAUs were benefited by these workshops.

Management of Agricultural Education 29-10-98
31-10-98

A three-day workshop on Management of Agricultural Education was organized to deal with the management problems of Agricultural Education and the best management solutions for taking the Agricultural Education to the 21st Century. Various topics covered in the workshop included Management of agricultural education perceptions for 21st Century, Addressing curriculum modification for eco-sustainability, Management of change in Agricultural Universities, and Time management and leadership issues.

Thirty Deans from various State Agricultural Universities participated in the workshop, co-ordinated by Dr S.N. Saha and Dr A. Gopalam.

Brainstorming workshop on Home Science Education 02-12-98
03-12-98

A two-day Brainstorming workshop on Home Science Education was organized to provide a forum for discussing various issues related to strengthening Home Science education and to evolve a new curriculum for increasing the effectiveness of Home Science education.

Vice Chancellors, Deans, and senior faculty members of various Universities, numbering 37, attended the workshop. Dr Carol Meeks, Dean, College of Family and Consumers Sciences, Iowa State University, USA, and Dr S.L. Mehta, Deputy Director General (Education), ICAR, New Delhi, participated in the



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Rajbhasha 28-12-98
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programme as resource persons. Dr D. Rama Rao and Dr N. Sandhya Shenoy co-ordinated the workshop.

A two-day workshop on Rajbhasha Karyanvayan Mein Anubhavjanya Prayog was organized to discuss about the experiential usage of Hindi in various ICAR institutes and to identify practical solutions to the problems faced in the implementation of official language.

Assistant Directors, Technical Officers, Translators, and Scientists-in-charges of Hindi Cells, working in various ICAR institutes, numbering 42, participated in the workshop, organized by the Hindi Section of the Academy.

Deans work- 18-03-99
shop on 20-03-99
Manage-
ment
of Agricul-
tural
Education

The workshop essentially deliberated on issues related to management of agricultural education for meeting the needs of the 21st century. During the workshop, the participating Deans were apprised of the recommendations of Dr M.S. Swaminathan Committee on Agricultural Education, which facilitated the participants to discuss various issues related to management of agricultural education. In addition, various other aspects related to time



management, leadership issues, organizational behaviour, and journals and databases were also covered during the workshop. The Deans were exposed to an innovative concept of workshop within a workshop. The faculty members of M.S. Swaminathan Research Foundation dealt with the modification of Agricultural Education in order to integrate the issues relating to gender equity. The participating Deans presented

the action plans for their respective Universities, which were then discussed for necessary implementation.

The programme had an active participation of 26 Deans from various State Agricultural Universities.

Meetings

Project Advisory Committee Meet 04-08-99

The fourth Project Advisory Committee (PAC) meeting of the DST Project "Professional Women in Agriculture" was organized at the Academy. The committee reviewed the progress of the project and recommended necessary future course of action.

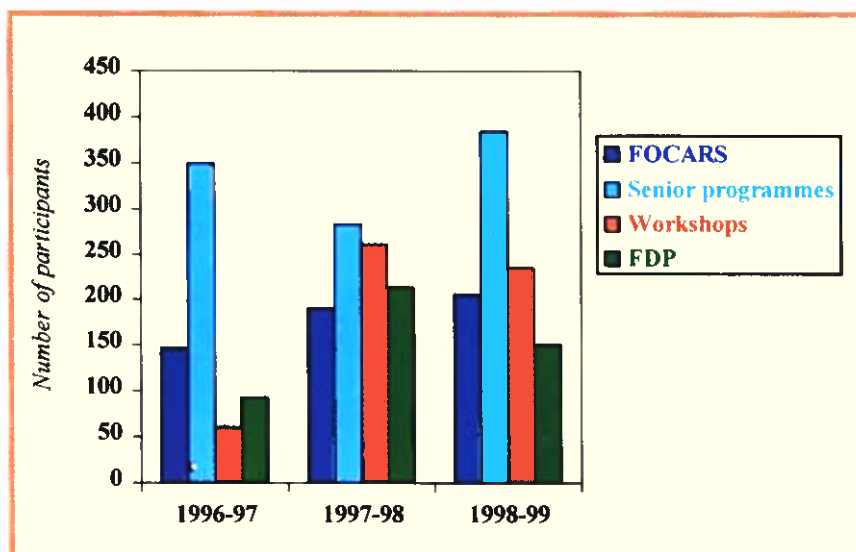
Advisory Group Meet of the Project on Distance Training for Agricultural Research Management 18-02-99

The first meeting of the Advisory Group of the Research Project "Distance Training for Research Management", which

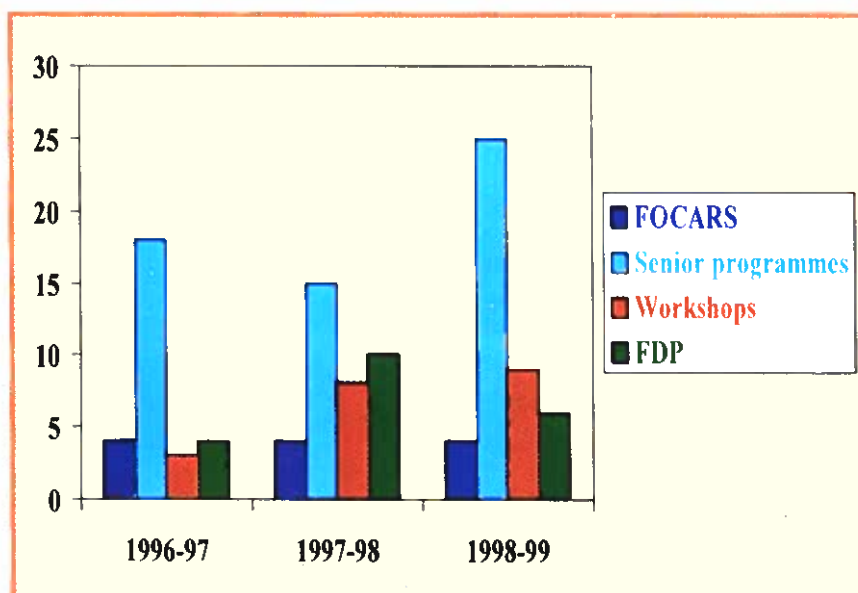


comes under the Work Plan of the agreement between ICAR and ISNAR, was organized at the Academy, under the Chairmanship of Dr S.L. Mehta, Deputy Director General (Education), ICAR, New Delhi.

Participation in programmes during the last three years



Number of programmes organized during the last three years



Feedback

The Academy organizes various types of training programmes, workshops, and seminars for the senior-level scientists and executives of the National Agricultural Research System, in addition to the Foundation Courses offered to the scientists newly recruited into Agricultural Research Service. On an average, about 40 programmes are organized every year. Whereas some of these programmes are of a recurrent and regular nature, others are organized based on the needs and demands of the client organizations. One of the striking features of the programmes organized by the Academy is the feedback on various aspects of the programmes, gathered through a structured questionnaire. The feedback is used constructively to improve the future programmes organized by the Academy. The programmes of the Academy are, thus, dynamic in nature. The strengths and weaknesses of the major programmes, organized during the period under report, as brought out through the feedback, are briefly presented below.

Programme	Feedback	
	Evaluative	Corrective
FOCARS	* Extensive practice with computers * Commitment and punctuality of conduct * Skills in project management	* Guest lecture and institutional visit need to be increased * More time needed for coverage on project proposal preparation * FET team composition needs re-examination
Senior-level Programmes	* Acquisition of new knowledge and skills * Practice sessions and participatory mode of learning * Good educational experience	* More training resource materials may be provided * Need for refresher courses and follow-up programmes

Sponsored Programmes

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| <ul style="list-style-type: none"> * Relevance and need of the programme * Commitment and expertise of the faculty members * Informal and non-formal learning from the programmes | <ul style="list-style-type: none"> * Visits to other institutes inadequate * Tighter schedule of the programme * Non-supply of training resource material of guest speakers |
|--|--|

Workshops

- | | |
|---|--|
| <ul style="list-style-type: none"> * Usefulness of group discussion and exercises * Useful training material provided * Provided a forum for sharing information on topical issues | <ul style="list-style-type: none"> * More case studies need to be used * Inadequate representation of Directors and top-level management functionaries in programmes |
|---|--|

Research

On-going Research Projects

Apart from the various training activities, the faculty members of the Academy are engaged in conducting research in important areas of agricultural research and education management. The outcome from research provides resource materials for conducting training which, in turn, increases the credibility of the training programmes. The progress of the various on-going research projects is briefly described below.

- **Research Project Management in the National Agricultural Research System**
(P. Manikandan, T. Balaguru, and S.N. Saha)

The project envisages to collect information, through questionnaire and also through personal interview, on various aspects related to research project management in NARS. The information is planned to be collected from different people, who include Project Leaders, Heads of Divisions, In-charges of Technical Cell, and Heads of Institutions. A detailed questionnaire for collecting the necessary information from the in-charges of the Technical Cell has already been developed. The questionnaire has two parts, one for general information and the other for technical information. In the technical information, 33 questions have been framed to collect information on five major areas, which are Proposal of research projects, Monitoring and evaluation, Research information management, Additional functions of the technical cell, and General issues. The questionnaire contains multiple choice-type questions as well as open-ended questions. The questionnaire is expected to generate valuable information on many aspects proposed to be studied through the project. Suitable questionnaires for others are under preparation.

- **Role of NGO and GO KVKs in Agricultural Development with special reference to Andhra Pradesh**
(N. Sandhya Shenoy and K. Vidyasagar Rao)

The project aims to identify the role of NGO and GO KVKs in agricultural development, to study the expectations of clientele groups from KVKs, and to identify the factors affecting the performance and operational constraints of KVKs. A questionnaire was developed and tested. The questionnaire was finalized based on the pre-test. The data collection from KVKs is in progress.

- **Social Auditing of Agricultural Programmes: Case Studies**
(S.K. Nanda and B.S. Chandel)

The conceptual framework, necessary for the progress of the project, was developed.

- **Implementation of Farming Systems Research (FSR) in India : Case Studies**
(T. Balaguru and P. Manikandan)

The project aims at developing case studies, with the objectives to investigate the conduct

and management of FSR in India, and to evolve suitable strategies for effective implementation. After a thorough literature search, a comprehensive questionnaire was developed and revised by interacting with the practitioners of FSR in the Indian NARS. The data collection is in progress.

● **Interface between Agricultural Research, Development, and Entrepreneurship**
(R. Kalpana Sastry and B.S. Chandel)

The project was initiated in December 1997. The review of literature has been completed.

● **Technological Forecasting of Future Oilseeds Scenario in India**
(D. Rama Rao, R. Kalpana Sastry, and V. R. Kiresur)

The project, supported by AP Cess Fund scheme, was initiated on April 1, 1997, and is expected to be completed by March 31, 2000.

The following entities of oilseeds have been forecasted for 2010 AD. They are: area, yield, consumption, competing crops, constraints of dryland research, specific end uses, influencing sectors, research breakthroughs, limitations on TOT, reasons for oilseed cultivation, seed supply and state role, blending, international trade, processing improvements, and government policies. Some salient findings and strategies, brought out based on the forecasts, are given below.

- ┐ The forecasted area under total oilseeds indicated an increase from 26 million hectares of the present level to 32 million hectares by 2010 AD, whereas the average yield showed an increase from 900 kg/ha of the present to 1150 kg/ha by 2010 AD. Similarly, the per capita oil consumption would increase from 8 kg/annum to 10.3 kg/annum by 2010 AD.
- ┐ The major competing crops for different oilseeds by 2010 AD would be: paddy, sunflower, and millets for groundnut; wheat and millets for sunflower; wheat for rapeseed and mustard; sunflower and gram for safflower; paddy for soybean; groundnut, millets, and pulses for sesame; and millets for castor.
- ┐ Slow results of the dryland research, insufficient resources, uncertain rainfall, lack of access to germplasm, lack of appreciation for basic research, and lack of interest in researchers were observed as the major factors affecting dryland oilseed research.
- ┐ The time required to evolve oilseed varieties for specific end uses are: for confectionery type in both groundnut and sunflower within seven years, low erucic acid in rapeseed and mustard in five years, spineless variety of safflower in five years, bold white seed in sesame in five years, and low linoleic acid and double purpose linseed in about 11 and eight years, respectively.
- ┐ The sub-systems associated with oilseeds that are likely to have more impact on overall oilseed situation by 2010 AD, in the order of importance, are: TOT (development), research and education, marketing and trade, and processing.

- ❑ Some major research breakthroughs that are possible in the next ten years in each of the oilseeds are: development of drought tolerant varieties, development of HPS variety for export in groundnut; improvement in seed viability, and herbicide application technology in soybean; development of photo insensitive varieties with low senescence, exploitation of germplasm for HYVs in sesame; improved techniques to augment higher percentage of pure seed, development of alternaria-resistant varieties in sunflower; deregulation in use of TM hybridization technology and breeding alternaria resistant varieties in rapeseed and mustard; identification of cytoplasmic genetic male sterility system and standardization of hybrid seed production technology in safflower; good composites and agro-production technologies under marginal lands in niger; early duration hybrids and resistant varieties to lepidopterous pests in castor; and genetic manipulation to make oil edible and techniques for value addition for commercialization in linseed.
- ❑ Low returns, unreliable and poor markets, inadequate and untimely supply of inputs, and inadequate incentives for effective technology transfer were cited as the major limitations for proper transfer of oilseed cultivation technologies to farmers.
- ❑ Medium farmers are likely to give more preference for oilseed crops cultivation. The reasons for their preference being: commercial nature of crop, crop rotation, exposure to improved technologies, and local market demand.
- ❑ There is a need to increase the seed replacement rate from the present 15-30% to a desired level of 50-60% for different oilseeds. To achieve this, the state agriculture departments have to take a major lead in supply of seed to farmers and priority finance for seed multiplication.
- ❑ The present market share of less than 5% for blended edible oils in total edible oil market is expected to increase to 8% by 2010 AD. The basis for blending of oils in future would be health, cost, keeping quality, and taste, in that order.
- ❑ Among the oils, groundnut, rapeseed and mustard, safflower, niger, castor, and linseed are expected to be cheaper in domestic market by 2010 AD, whereas sunflower, soybean, sesame, and oilpalm will be cheaper in international market.
- ❑ The possible technical improvements in oilseeds processing that could be envisaged by 2010 AD are increased use of solvent extraction and production of pesticide-free byproducts.
- ❑ The government policies on futures trading, liberalization of seed import, efficient in-land handling and transport, stock decontrol, and development of infrastructural facilities are expected to overcome constraints faced by processing units in the next ten years.

● **Studies on Application of Indigenous Technical Knowledge (ITK) in Animal Management for Enhancing Production**

(P. Kumar , K. Vidyasagar Rao, and N. Sandhya Shenoy)

A questionnaire has been developed to collect the required information from the farming community. The data collection work is in progress.

● **Evolving a Suitable Methodological Framework for Priority Setting in Agriculture**
(T. Balaguru)

The project was initiated with the objective of developing suitable methodology for setting research priorities in agriculture. The data collection is in progress.

● **Impact of NAARM Training in Agricultural Research Project Management on ICAR Scientists and Assessment of Their Training Needs**
(K. Vidyasagar Rao and N. Sandhya Shenoy)

NAARM has been organizing training programmes in Agricultural Research Project Management to ARS Probationers and to other senior scientists since 1976. The opinion of the participants and the impact of these programmes are very much essential to restructure and modify them in a meaningful way. A project is, therefore, undertaken to study the impact of NAARM training.

The present study covers the training programmes conducted during 1992-95. The data from the participants, who attended various training programmes during this period, were collected through a structured questionnaire. The data compilation, analysis, and the draft report are over. The final report is under preparation.

● **Distance Training for Agricultural Research Management**
(Nicole Brunet and Jagannadham Challa)

This project is a collaborative venture of ISNAR and NAARM. During the period under report, three background papers were prepared which would facilitate identification and understanding of various issues needed for the progress of the project. The first Advisory Group Meeting of the project was held during this period. The institutions identified for the study are Mahatma Phule Krishi Vidyapeet, Rahuri; Acharya N.G. Ranga Agricultural University, Hyderabad; University of Agricultural Sciences, Dharwad; and Directorate of Rice Research, Hyderabad. During the period under report, survey questionnaires were developed for Agricultural research management training needs assessment, Institution communication and training facilities, Distance training institutional assessment, and Management training capacity.

● **Performance Assessment and Accountability Enhancement of Agricultural Research**
(T. Balaguru, K.P.C. Rao, and R. Kalpana Sastry)

As a part of the Rs.5.3 crore sub-project on Human Resource Development for Agricultural Research and Education, this proposal was developed jointly by NAARM and ISNAR, as one of the O & M initiatives undertaken by NATP. At an initial meeting, in which NAARM faculty (project team members) and ISNAR scientist (Govert Gijssbers) participated, the basic framework for implementing the project was discussed and some of the important publications on the topic were exchanged among the team members.

● **HRD Strategies for Organizational Effectiveness in NARS**
(Jagannadham Challa and K. Vidyasagar Rao)

The project is proposed with the following objectives.

- * To develop management systems approach for enhancement of organizational effectiveness through the system of Management by Objectives in five ICAR institutions including one Deemed University
- * To develop and standardize case models for institutions for research productivity and for personal development of individuals
- * To enhance the co-ordinated efforts of research managers and to establish norms for identifying the contribution of individuals

The project is of 2 years duration and is funded from AP Cess Fund Scheme. The institutes to be considered for the proposed study are CIFE, Mumbai; CRIDA, Hyderabad; CRRI, Cuttack; IIHR, Bangalore; and IVRI, Izatnagar.

Concluded Research Projects

● **Professional Women in Agriculture**
(D. Rama Rao and N. Sandhya Shenoy)

The project was initiated on November 1, 1996, and was terminated on August 31, 1998. The project aimed at studying the status and growth of agricultural women graduates, career opportunities for agricultural women graduates, and to identify issues pertaining to women professionals and the factors influencing them.

The target population included the UG, PG, and Ph. D degree holders in agriculture and allied sciences, i.e. all disciplines offered by agricultural universities. The number of female students, with professional degree in agriculture, has increased many fold, i.e. from 387 (3.9% of total) in 1980 to about 3,773 (21.6% of total) in 1996.

The employers of professional agricultural women such as ICAR, agricultural universities, banks, state and central departments, private sector, the unemployed women graduates, and students were contacted for the survey purpose. Four different questionnaires were used - three for maintaining databases on agricultural graduate students, employers, and organizations employing them, and one for survey covering personal, academic, career information, career expectations and values as well as career problems.

The estimated number of agricultural women graduates in the country in 1995 was 14,460. Purposive sampling was done for the survey using the snow-ball technique. The survey could identify 7,821 graduate women, out of which 1,443 expressed their willingness to participate in the survey. The final sample consisted of a total of 1,298 respondents, out of which 454 were male and 844 were female. Male respondents were included to identify gender-neutral issues and also for comparison of perceptual differences.

The project report shows the changes needed in procedures, rewards, and policies of both government and private agencies. The work is the culmination of serious survey, case studies, and brainstorming workshop.

Based on the findings it can be concluded that there is no major perceptual difference between women and men professionals as both regard work as gender-neutral and give more importance to competence. The results are discussed in terms of per cent respondents agreeing to or disagreeing to a particular issue or event. Broad subject-wise findings are given in the following.

Agricultural education

A study of the school background of the female respondents indicated that 14.9% were from rural and 85.1 per cent were from urban schools. The professional education in agriculture was opted as the second best (next to medicine) by female respondents (18.5%) at the time of entry into college. Agriculture was preferred as it was regarded as a challenging profession working with nature, tapping the potential resources for the welfare of the country. It is worth noting at this juncture that parents had been most influential in motivating to take up education in agriculture, more so, the mother in case of female respondents. In agricultural education, field work is mandatory, and the study revealed that majority of the respondents had not faced difficulties in field work. However, 12 per cent of the respondents claimed to have faced difficulties in field work due to one of the following reasons: lack of practical training and guidance, physical strain, non-agricultural background, negative attitude of colleagues, and distance from field. Student respondents opined that, HomeScience (40.9%), Horticulture (18.9%), and Extension (15.7%) specializations were the convenient specializations for graduate women.

Employment issues

Data on job-status of female graduates indicated that 72.6 per cent were employed, 6.6 per cent were unemployed, and 20.8 per cent were students. The situation of unemployment in agriculture was low as compared to other professions. Female respondents, who were unemployed, cited the reasons that employment in agriculture involved frequent travel and intensive field work. As far as the employing organizations are concerned, it was seen that the largest number (39.7%) of female graduates were employed in educational institutions. This was followed by government development departments (19.2%), research institutions (7.9%), private sector (4.2%), and financial institutions (1.1%).

The study pointed the fact that sexual harassment faced by the female respondents was low at the place of work and about 22.3 per cent indicated facing this problem. Thus, a congenial atmosphere / work condition existed for women at the place of work, in most organizations.

Career perceptions

Majority of women (57.2%) opined that they had to do better than men to get equal professional recognition. About 52.8 per cent women revealed that family is of primary concern when compared to career. Both, male (82.8%) and female (85.6%) respondents, considered

family support a key factor to a successful career. As infrastructure facilities are poor in field stations, women (35.5%) felt that it would be convenient to work in such stations during the initial entry of their career when the family obligations would be minimum.

In the choice of colleagues at work, majority of the respondents opined that work is gender neutral. In the choice of their supervisor, 61.5 per cent of the female respondents indicated gender-neutrality, 22.9 per cent preferred male supervisor, and 15.6 per cent preferred female supervisor. Data on preferences for choice of subordinate revealed that 66.3 per cent of the female respondents implied gender-neutrality, 16.5 per cent preferred male subordinates, and 17.2 per cent preferred female subordinates.

Management issues

The non-availability of physical facilities like creche, school, wash room, accommodation, transport, and health facilities were reported by majority of the respondents. It is to be noted as a vital issue that both female and male respondents expressed concern on absence of flexible working hours. Gender prejudice was cited as an important reason by female respondents for not getting job, especially in private sector. The awareness and participation in employee associations and training programmes was relatively low in case of female respondents. Some of them (27.0%) had not undergone any training programme in their professional career. Low salary, distance of work place from residence, and lack of accommodation were cited as reasons for refusal of job by women.

On the question of reservations to women in education and employment, nearly one third female respondents were against any reservations. However, two third female respondents felt the need for reservations in all activities supported by government.

The number of female respondents with top level management positions was negligible. To alleviate some of the managerial problems, the female respondents felt the need to associate women in all decision-making bodies like selection committees, etc.

Recommendations for Action

Based on the analysis of nation-wide survey findings, and supported by case studies and brainstorming workshop, the following recommendations are made for alleviating the problems faced by women graduates and also to encourage more and more women graduates to come to mainstream economy.

Educational issues

- * As urban graduates are reluctant to serve in rural areas, more number of rural students be admitted in the agricultural courses so as to increase the number of professional graduates with rural background, who may take up extension-related jobs/ activities.

- * Women reservations are to be extended to all activities supported by state such as scholarships, admissions, nominations to awards, curricular and co-curricular activities.

(Action : SAUs and ICAR)

Technology transfer issues

- * The organizational climate in the line departments needs to be made gender- friendly to encourage women to take up extension jobs. The women employees are to be provided necessary facilities and services like transport, accommodation, and spouse employment.

(Action : Development Departments)

General issues of working women

- * It must be made mandatory to associate at least one woman member in all committees such as selection panels for recruitment/ promotion, grievances, etc. This may minimize the discrimination met by certain women, purely on gender basis.
- * For the better satisfaction of employees, work structure can be modified incorporating features like flexible timings, easy leave option, and contract jobs.
- * Personnel policies to be evolved aiming at spouse employment and preferential placement for working women.
- * For out-station tours, availability of facilities like transport and accommodation may be made a prerequisite.
- * Essential minimum infrastructure facilities like board, lodging, transport, toilets, etc., exclusive for women, be provided at training institutes and work places.
- * HRD programmes for professional women in agriculture need proper planning in timing and duration, keeping in view the dual responsibility of women employees.
- * More women professionals should be encouraged to become members in professional societies and proportionate opportunities be given to participate in conferences and training programmes.

(Action : All employing organizations)

There is a need for a co-ordination body (women cell) and also a need to network agricultural professional women at organizational and national level. National Commission for Women (NCW) may provide broad guidelines and insist for creation of a Women Cell in each organization for implementation of women-related policies. It may establish an organization-independent mechanism to review the progress from time to time.

(Action : NCW)

Consultancy

Consultancy is one of the important activities of the Academy. Depending upon the specific invitations from client organizations and by virtue of the expertise available, consultancy services are undertaken in some of the mandated areas of the Academy. In the past, consultancy services were provided in various areas for a number of organizations. The following consultancy services were provided during the year under report.

Consultancy offered to M.S. Swaminathan Research Foundation, Chennai

At the request of M.S. Swaminathan Research Foundation, Chennai, a short-term consultancy was provided from August 4 to September 3, 1998. Dr T. Balaguru, Principal Scientist, provided consultancy support to review and consolidate the achievements made by the UNDP-supported Biovillage Demonstration Project, being implemented by M.S. Swaminathan Research Foundation, in 19 villages, in the Union Territory of Pondicherry. To take stock of the achievements made by the project, operating in 19 villages, various kinds of information from primary and secondary sources were collected, collated, and analysed for drawing managerial inferences. Besides presenting the results in a readily usable form, various technical, managerial, and policy-level suggestions were made to enhance the effectiveness of the project in the remaining period of its implementation.

Consultancy offered to the Government of Tamil Nadu

Dr R. Chithiraichelvan, Senior Scientist, acted as a consultant and faculty for the State-level Course on Video Production, in which officials of the Department of Agriculture, Government of Tamil Nadu, participated. The programme was organized from November 23 to 27, 1998, at the Directorate of Agricultural Extension, Tamil Nadu Agricultural University, Coimbatore. The programme was funded and sponsored by the Ministry of Agriculture of the Government of India.

Centralized Services

A number of Centralized Services, established at NAARM, extend various services and facilities for the effective implementation of the different activities undertaken by the Academy. Major developments brought about in some of the service units of the Academy, during the period under report, are briefly presented.

The Computer Section and ARIS Cell witnessed rapid development, during the period under report, in terms of new hardware and software installation, Internet connectivity, and skills development.

A total of 30 new P - II computers and over 50 P - 60 and 486 computers were installed with a view to providing computer access to faculty members, staff, and participants of various training programmes. AC - band VSAT, through ERNET, was installed and later repointed for persistent Internet connectivity for the Academy. With the installation of VSAT, the Academy acquired continuous full TCP/IP connectivity. Optical Fibre Networking, with 155 MBPS capability, for the entire Academy, was undertaken during this period.

With Internet connectivity, the Academy, with its own domain name (naarm.ernet.in), installed new software for a multi-platform network with Norell Intranetware, Windows NT, Sun OS, and Linux, as operating systems, providing different services like Internet gateway, Proxy, individual e-mail, DNS, and DHCP. Netscape Suitespot was installed to provide basic Internet and Intranet services. NAARM revamped its Intranet structure, content, and organization, with the addition of the new capabilities. The NAARM Website was integrated with the nascent Global Learning Centre being developed at NAARM. Staff skills were also enhanced to meet the additional responsibilities of Intranet and Internet Web, and Ftp site management.

The Video Instruction and Production Laboratory (VIPL) of the Academy is one of the best facilities available in the NARS. The existing facilities were augmented and additional equipment were procured during the period under report.

Two new systems were procured - one for adding titles to videos in Hindi and the other for animation. The non-linear Edit Station from Sony was put into use and editing of the videos, using this system, has been standardized. Sixty five video programmes were produced by trainees of Foundation Courses and other senior programmes, who had extensive hands-on experience in video production. The existing video on "NAARM - An Introduction" was modified with new sequences and by inclusion of fresh rushes.

The Academy caters to the requests received from SAUs and ICAR institutes for the supply of copies of video programmes produced by the Academy. During the period under

report, 87 cassettes were sent to various ICAR institutes and SAUs, based on the requests received. This had generated an income of about Rs.32,000/- for the Academy. An additional income of Rs.5,00,000/- was generated for the services to be rendered for producing two video films for CPCRI, Kasaragod.

Information Centre

The Academy has established a well-organized Information Centre with a rich collection of books and journals of diverse areas, which include Agricultural Research Management, Project Management, Educational Technology, Educational Psychology, Computers, Information Technology, Management and Communication, Human Resource Development, etc. During the year under report, 1000 books on various topics were added besides subscribing to 150 foreign journals and 150 Indian journals. In addition, the Centre has a good collection of reports and other publications of various international organizations such as UNESCO, FAO, World Bank, and CGIAR institutions. The Centre extends facilities for information search through CD-ROMs. During the period under consideration, the Centre has automated the activities of the library and databases of books and journals were developed on CDS/ISIS. The library also subscribes to databases on CD-ROMs for AGRIS (1989-99), AGRICOLA (1980-94), and Current Contents on floppy diskettes (1997-99).

Scientific information management is an integral part of almost all the training programmes organized by the Academy. The Information Centre is actively involved in imparting training on the on-line search and browsing and downloading the information through Internet, to the trainees of various programmes. At present, the Centre has three Pentiums, two of which are connected with Internet.

Hostel

During the period under report, a new block has been added to the Halls of Residence. This new block has 16 double-bedded rooms and 16 single rooms. The kitchen of the Hostel is further equipped with the addition of a new cooking range and new cutlery. With these improvements brought about in the facilities, the Hostel Unit is able to extend better services to a large number of trainees and scientists who stay in the Halls of Residence and Scientist Home of the Academy.

Farm Section

The Farm section had taken up extensive tree planning during the period under report. Vansamrakshan programme was organized with a view to take up simple agronomic practices on the trees which were planted in 1997. This programme was conducted in a worker participatory mode on August 15, 1998, with the involvement of the entire staff of the institute including the trainees of the ongoing programmes on that day. The samrakshan programme

was a success and all the plants planted in 1997 are surviving. A block was developed where, as per the management committee recommendations, trees are to be planted by the VIPs coming to the Academy. During the year under consideration, Management Committee members, including the Chairman, external auditors, and other VIPs had planted trees in this block.

New plantations of mangoes, citrus, sweet oranges, and teak were laid out in various blocks. In a new initiative, four perennial but value added product-based plant species were introduced as a part of dryland orchards. The plants were improved varieties of curryleaf (var : Suhasini), Annato (*Bixa orellana*) - a dye plant, drumstick (var : Dhanjar), and Henna.

During the period under consideration, the 50 years of India's Independence, ICAR logo, and letters ICAR and NAARM were developed and planted with various plants. The development of such logos was done for the first time, and with great enthusiasm by all the farm personnel.

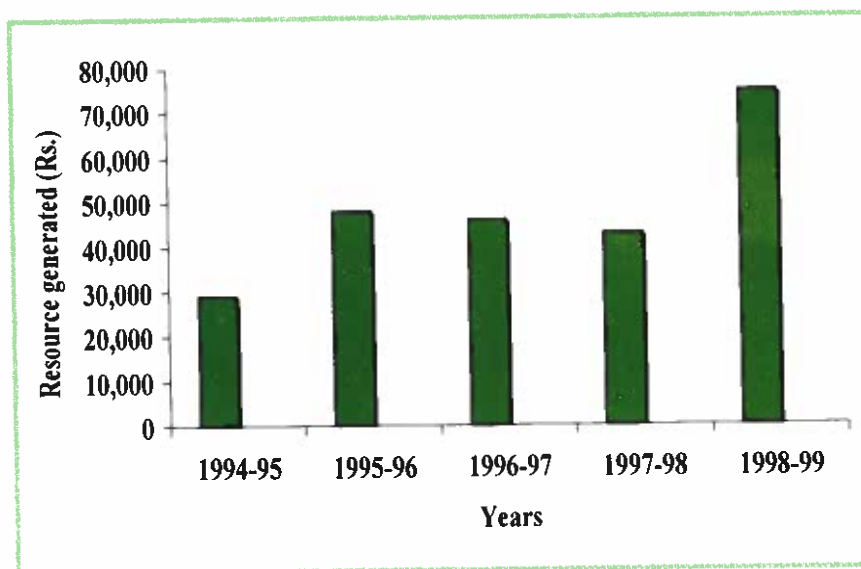


The entire farm section worked as a team and participated in different exhibitions in the twin cities during the period. The various prizes won by the Academy are:

- a) Andhra Pradesh Formation Day celebrations at Public Gardens (November 1 - 2, 1998) - One first prize and two second prizes.
- b) Rose Show conducted by the Hyderabad Rose Society at Public Gardens (December 11 - 12, 1998): Rolling Trophies - 05, First Prizes - 06, Second Prizes - 08. These included the Best Indian Rose and Collection of Best White Rose awards.
- c) Rose Show conducted by the Secunderabad Horticulture Society at Contonment Gardens, Secunderabad (December 18 - 19, 1998): Rolling Trophies - 07, First Prizes - 16, Second Prizes - 18. These included Best Red Rose, Indian Rose, Best Institutional Garden, Rosarian of the year, and Best Yellow Rose Collection awards.
- e) 59th All India Industrial Exhibition conducted at Exhibition Grounds (February 5 - 6, 1999): First Prizes - 04, Second Prizes - 06. These are in addition to the Queen of the show and Best fragrant rose award.

The farm has been able to generate an income of Rs.74,528/- , through various activities taken up during the year under report. There has been a substantial increase in the amount generated from farm, as compared to the previous years.

Resource generation from farm activities during the last five years



Other Strength-building Activities

Agricultural Human Resource Development Project

Agricultural Human Resource Development project (AHRD) at the Academy supports primarily the activities on improving the quality of agricultural education. NAARM being the only training Academy in the agricultural research and education management area, several State Agricultural Universities (SAUs), ICAR institutes, and other institutions utilize the expertise available with the Academy to get their manpower trained in the broad areas of Agricultural Research Management, Educational Technology, Educational Psychology, and Instructional Effectiveness. Tamil Nadu Agricultural University, Coimbatore; Tamil Nadu Veterinary and Animal Sciences University, Chennai; and Acharya N.G. Ranga Agricultural University, Hyderabad, deputed their faculty members to NAARM to undergo training under AHRD funds, in various specific areas. The objectives of these training programmes were:

- to train the teachers in agricultural education management
- to strengthen the educational initiatives of teachers for optimizing learning-teaching
- to bring about behavioural modifications by organizing training in attitude-related issues.

NAARM - ISNAR Research Project on Distance Training for Agricultural Research Management

A project on Distance Training for Agricultural Research Management is jointly undertaken by NAARM and the ISNAR at the Netherlands, with special assignments for Wye College External Programme (WEP) in the UK, and the Canada-based Commonwealth of Learning (COL).

The project aims to develop a sensible distance training option to reach the unreached and to provide in-service training opportunities in a more non-formal and open-ended manner, with easy access to information and self-learning materials. The goal of the research project is to enhance the contribution of NARS for sustainable rural development, leading to the alleviation of rural poverty in India through distance training of researchers and research managers of regional research stations located in rural areas.

The project, scheduled for three years starting from October 1998, and located at NAARM, Hyderabad, is planned to be carried out on action research methodology consisting of three components such as applied research, adaptive research, and pilot distance training programme. Dr Nicole Brunet of ISNAR is the Project Manager and Dr Jagannadham Challa, Principal Scientist, NAARM, is the Principal Investigator of the project. The project is executed in collaboration with Wye College External Programme (WEP), UK, and Commonwealth of Learning (COL), Canada. The Project Advisory Group, chaired by Dr S.L. Mehta, Deputy Director General (Education), ICAR, New Delhi, shall function as a group of peers for providing guidance throughout the project duration.

National Agricultural Technology Project

Under the aegis of the World Bank-supported National Agricultural Technology Project (NATP), NAARM jointly developed a project with NCAP and IASRI, entitled 'Institutionalization of Priority Setting, Monitoring and Evaluation, Networking of Social Scientists, and Human Resource Development for Agricultural Research and Education Management'. The Academy is specifically entrusted with the Human Resource Development component of the project. Out of the total project cost of Rs. 9.54 crores, NAARM is provided with a budget of Rs.5.09 crores for five years period from 1998-99 to 2002-03. During the next five years, the Academy will be concentrating on various training activities, which include training senior executives of ICAR, Management Development Programme for newly recruited research managers of ICAR and SAUs, and training scientists of NARS on research priority setting in agriculture. Under research studies, the projects undertaken include Performance assessment and accountability enhancement of agricultural research organizations in India, Forecasting trained agricultural manpower, Developing computer-mediated courseware for facilitating learning-teaching, and Capacity building of NARS in decision-making on natural resource management. A team of faculty members have been identified to undertake each of these activities.

Foreign Visits

* Dr J.C. Katyal, Director, NAARM, and Dr Jagannadham Challa, Principal Scientist, NAARM, attended the two-day Planning Meeting of the Project on Distance Learning in Agricultural Research Management in Indian NARS, held on June 11 and 12, 1998, at ISNAR Headquarters at the Hague, the Netherlands.

* Agricultural Human Resource Development (AHRD) Project at NAARM focuses on development of Educational Cells, strengthening of existing facilities at NAARM, faculty development by overseas training and study visits, and conducting teacher-improvement training programmes for the faculty members of SAUs for improving the quality of teaching and learning in the Agricultural Universities. Under the AHRD Project, Dr A. Gopalam, Principal Scientist, was deputed to undertake a study tour to Wye College External Programme, University of London, U.K., from August 9 to 29, 1998. The study tour was undertaken to conceptualize the intricate details of distance learning for adaptability and to develop insights into distance learning by one to one interaction.

Faculty News

* Dr S.N. Saha, Joint Director, attended a training programme on "Leadership and Society - An Integrated Approach to Knowledge and Information", conducted by the National Institute of Advanced Studies, at Bangalore, from January 4 to 23, 1999.

* Dr P. Manikandan, Principal Scientist, was awarded Postgraduate Diploma in Management (PGDIM) by the Indira Gandhi National Open University (IGNOU), New Delhi.

Dr M.M. Anwer, Principal Scientist, was awarded Postgraduate Diploma in Human Resource Management (PGDHRM) by the Indira Gandhi National Open University (IGNOU), New Delhi

NAARM Hosted the ICAR (Zone III) Inter-institutional Tournaments

The 15th ICAR Inter-institutional Tournaments (Zone III) was hosted by the Academy and was conducted at Lal Bahadur Stadium, Hyderabad, from December 15 to 18, 1998. A total of 506 sports personnel, including 59 women, from 16 ICAR institutes participated in the tournaments. Inaugurating the tournaments, Shri



K.V. Ramana Chari, Director of Sports and Youth Services, Govt. of Andhra Pradesh, Hyderabad, said that one must have the determination and resolve to see the things sportively. Difficulties and obstacles may crop up during the course of an attempt, but they can be overcome by a strong willpower. As the difficulties and disappointments are overcome, hope and encouragement will arise, and these will be helpful to get over further difficulties, he said.

Dr J.C. Katyal, Director, NAARM, and Chairman, Sports Meet, presided over the function. He said that the very idea of tournaments is to develop discipline, co-operation, understanding, and fraternity among the various functionaries of ICAR institutes. Sporting events of this kind will be of great help in relieving tensions arising from the routine office and laboratory work. These will help cultivate a healthy and happy working environment, he said. In the long run, sports promote friendship and fraternity, and help to make the life of the people healthier and happier, he added. Earlier Dr S.N. Saha, Joint Director, NAARM, and Convener of the tournaments, welcomed the players.

The NAARM sports contingent, consisting of 38 members, which included four ladies, was led by the Sports Manager Dr A. Debnath. Ms Rukmini Ammal brought laurels to the Academy by winning four prizes, three in individual events and one in group event. She bagged first prize each in shot put and discus, and third prize in javelin throw. She also won runners-up prize in shuttle badminton (doubles) along with Ms Aneeka. Ms Aneeka also won the first prize in javelin throw. In the men's events, Mr M.K. Samson bagged second and third prize, in shot put and javelin throw respectively. Dr A. Debnath won the second prize in javelin throw. In the team event for men, NAARM won runners-up prize in table tennis.

Mr K. Raghuraman, Zonal Manager, Central Bank of India, who was the chief guest at the closing ceremony, distributed prizes and trophies to the successful athletes. Dr M.M. Anwer, Organizing Secretary, proposed the vote of thanks.

In the planning and organization of the ICAR (Zone III) Tournaments, the Academy had set an example of excellent team work, commitment, and co-operation. It was a rewarding experience for every member of the NAARM family, who contributed to the success of the tournaments.

Hindi Fortnight Celebrations



To create awareness about the official language and to create an environment congenial for the use of Hindi for official purposes among the staff and officers of the Academy, Hindi Fortnight was celebrated from September 7 to 19, 1998. Various Hindi competitions were organized to mark this occasion. Dr J.C. Katyal, Director, NAARM, in his address on the closing-day ceremony, appealed to all the employees to work in Hindi and avail the incentives provided by the Government of India in

this connection. Dr S.N. Saha, Joint Director and Chairman of the Official Language Implementation Committee of the Academy, said that the culture of a nation is nurtured by the language of that nation, and Hindi is the language that connects the entire nation as majority of the people speak in Hindi. Earlier Mr D. Venkateswarlu, Technical Officer (Hindi), welcomed the gathering. Dr A. Gopalam, Officer-in-Charge (Hindi), presented a report on the achievements of the Hindi section. Ms J. Renuka, Assistant Director (Hindi), proposed the vote of thanks.

Pulse Polio Camp

NAARM Health Centre organized Pulse Polio Immunization Programme on June 6, July 10, and August 27, 1998, as a part of the Government campaign of polio eradication from our country. Many children below five years age group, residing in and around NAARM campus, were immunized under this programme.

Hepatitis "B" Vaccination Camp

Periodical Hepatitis "B" vaccination camps were organized for the benefit of the staff, their family members, and trainees of the Academy. Ten such camps were organized in 1998, and a total of 424 persons, including 192 children and 153 trainees, were vaccinated at a subsidized rate.

Blood group testing and blood sugar estimation are being routinely carried out at the Health Centre of the Academy. Blood grouping and Rh factor determination were performed for 170 persons during the period under report.

Human and Financial Resources

Human Resources

Category	Sanctioned Strength	Posts Filled (as on 31-3-99)	Vacant Positions
Scientific	43*	23	20
Technical	67	61	6
Administrative	47	47	-
Supporting	41	41	-

*Includes three RMP positions

During 1998, ICAR assigned specific subjects to the sanctioned scientific positions at NAARM. Subject-wise and post-wise approved categorization is given below.

Discipline	Cadre Strength			
	Scientist	Senior Scientist	Principal Scientist	Total
Agricultural Economics	-	2	1	3
Agricultural Extension	-	3	1	4
Agricultural Research Mgmt*	-	4	4	8
Agricultural Statistics	-	1	1	2
Computer Applications	-	2	2	4
Educational Psychology*	-	1	1	2
Educational Technology*	-	3	2	5
Human Resources Management*	-	4	1	5
Mass Communication*	-	1	-	1
Research Project Management	-	1	4	5
Rural Sociology*	-	-	1	1
Total	-	22	18	40

*Discipline does not comply with the Disciplines of Agricultural Research Service

Financial Resources

The budget allocation and expenditure during 1998-99 are given in the following table. Information presented in table below also provides details on finances available from other sources and revenue generated during the year under report.

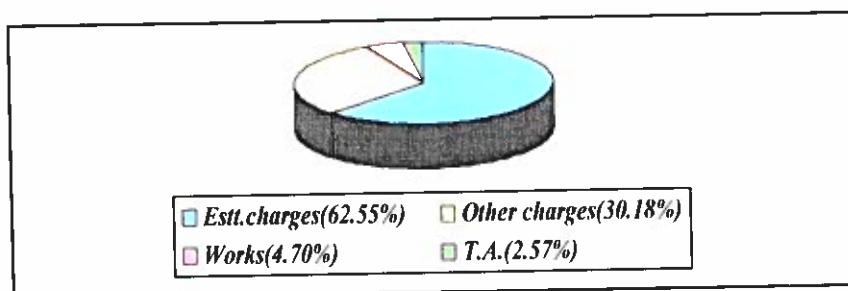
(Rs. in lakhs)

Scheme	Budget allocation	Expenditure
Non-plan		
□ Establishment Charges	300.00	218.86
□ Travelling Allowances	3.00	2.77
□ Other Charges	90.00	89.61
□ Works	--	--
Sub-Total	393.00	311.24
Plan		
□ Establishment Charges	22.00	20.56
□ Travelling Allowances	8.00	7.06
□ Other Charges	27.00	25.91
□ Works	20.00	17.99
Sub-Total	77.00	71.52
Other Sources		
□ Summer Institute	0.75	--
□ A.P.Cess	4.34	3.41
□ AHRD	102.59	83.91
□ NATP	40.00	15.18
□ PWA	2.79	2.51
Sub-Total	150.47	105.01
Grand Total	620.47	487.77

Resource Generation (Rs. in lakhs)

Revenue generated at the Academy	33.59
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Proportion of expenditure on various Heads (Plan + Non-plan)



Committees

Management Committee

The Academy is guided by the Management Committee in planning, organizing, and conducting its various programmes. The Management Committee approves various programmes and financial outlays thereupon. The Academy ensures their implementation, both in terms of physical targets and schedules, within the sanctioned budget heads. The Management Committee also supports and guides in formulation and consideration of proposals for Five Year Plan and Annual Plans, and takes up periodical review of progress of development schemes. During the year under report, one meeting of the Management Committee was held on December 5, 1998.

The composition of the Management Committee of NAARM is as follows :

Chairman

Dr J.C. Katyal, Director, NAARM

Members

Dr Anupam Verma, Dean, Indian Agricultural Research Institute, New Delhi
Mr Jannath Hussain, Secretary, Dept. of Agriculture, Govt. of A.P., Hyderabad
Dr D.N. Jha, Director, NCAP, IASRI Campus, New Delhi
Prof B.D.Joshi, R.K.College, Madhubani, Bihar
Dr S.Kannaiyan, Dean, Postgraduate Studies, TNAU, Coimbatore
Dr K.Krishnaiah, Director, DRR, Hyderabad
Dr S.L.Mehta, Deputy Director General (Education), ICAR, New Delhi
Dr Mruthyunjaya, Assistant Director General, ICAR, Krishi Bhavan, New Delhi
Mr B.Pandu Reddy, F & A.O., NAARM, Rajendranagar, Hyderabad
Ms Sabita Das, Ex-MP, Meharpur (P.O), Silchar District, Kachar, Assam
Dr I.V.Subba Rao, Vice-Chancellor, Acharya N.G.Ranga Agricultural University, Hyderabad
Capt. Mehar Singh, CAO, NAARM - Member Secretary

Official Language Implementation Committee

The Official Language Implementation Committee of the Academy provides periodical suggestions for implementation of Official Language Policy, and the rules and regulations of the Government of India that are issued from time to time. The Committee met two times during the period under report, the first meeting was on November 17, 1998, and the second one was on February 9, 1999. The Committee discussed several issues relating to the implementation of Hindi language during these meetings. The following is the constitution of the Official Language Implementation Committee.

Patron

Dr J.C. Katyal, Director, NAARM

Chairman

Dr S.N. Saha, Joint Director

Vice-Chairman

Dr A. Gopalam, Principal Scientist

Members

Dr Ajit P. Maru, Senior Scientist

Dr M.M. Anwer, Principal Scientist

Dr B.S. Chandel, Senior Scientist

Ms D. Dhana Laxmi, Library Assistant

Dr Jagannadham Challa, Principal Scientist

K. Janakiramaiah, Assistant Administrative Officer

Mr B. Pandu Reddy, Finance & Accounts Officer

Dr S. Shanmugam, Principal Scientist

Ms T. Vanisri, Stenographer

Mr D. Venkateswarlu, Technical Officer (Hindi) - Member Secretary

- A group of 48 final year B.Sc (Agric.) students of Kerala Agricultural University, College of Horticulture, Vellanikkara, visited the Academy on April 20, 1998.
- Jock R. Anderson, Evaluation Adviser, Agriculture and Human Resource Development Division, Operation Evaluation Department, Washington D.C., visited the Academy on April 23, 1998.
- Dr Paul Perrault, Programme Director, ISNAR, and Dr Nicole Brunet, Faculty, ISNAR, visited the Academy from April 27 to 29 for discussing with the Director and the faculty members about the possible collaboration with NAARM in the area of distance learning and creation of ICAR-ISNAR Regional Centre at NAARM.
- Six-member delegation from Iran visited the Academy on May 6, 1998, under the Indo-Iranian MOU signed for co-operation in agricultural research.
- A group of seventy four students of B.Sc. (Agricultural Marketing / Forestry) from University of Agricultural Sciences, Dharwad, visited the Academy on December 17 and 19, 1998.
- A group of twenty students of B.Sc. (Home Science) from Tamil Nadu Agricultural University, Home Science College, Madurai, visited the Academy on December 19, 1998.
- A group of 85 students of B.Sc. (Agric.), Tamil Nadu Agricultural University, Madurai, visited the Academy on December 29, 1998.
- A group of nine final year students of M.Sc. (Home Science Extension) from Sri Venkateswara University College, Tirupati, visited the Academy on November 18 and 19, 1998.
- A group of 80 final year B.Sc.(Agric.) students of Agricultural College, Tamil Nadu Agricultural University, Coimbatore, visited the Academy on October 22, 1998.
- A group of 40 final year B.Sc. (Agric.) students of Agricultural College, Tamil Nadu Agricultural University, Madurai, visited the Academy on October 23, 1998.
- A four-member delegation from Ethiopia visited the Academy on January 11, 1999, and discussed with the Director and faculty members.
- A group of 54 final year B.Sc. (Agric.) students of College of Agriculture, G.B.Pant University of Agriculture and Technology, Pantnagar, U.P., visited the Academy on January 2, 1999.

Visitors

- A group of 55 final year B.Sc. (Agric.) students of Pandit Jawaharlal Nehru College of Agriculture and Research Institute, Karaikal, visited the Academy on January 4, 1999.
- A group of 41 final year B.Sc. (Forestry) students of Dr Yashwant Singh Parmar University of Horticulture and Forestry, College of Forestry, Nauni, Solan, H.P., visited the Academy on January 4, 1999.
- Dr Stein Bie, Director General, ISNAR, visited the Academy on January 16, 1999, to discuss the modalities of operation of NAARM - ISNAR research project on Distance Training for Agricultural Research Management.



- A group of 27 final year B.Sc. (Agric.) students of College of Agriculture, Kerala Agricultural University, visited the Academy on February 10, 1999.
- A group of 60 final year B.Sc. (Agric.) students of Agricultural College and Research Institute, Coimbatore, Tamil Nadu Agricultural University, visited the Academy on February 16, 1999.
- A group of 58 final year B.Sc. (Agric.) students of College of Horticulture, Vellanikkara, Kerala Agricultural University, visited the Academy on February 22, 1999.
- A group of 40 final year B.Sc. (Agric.) students of Agricultural College and Research Institute, Coimbatore, Tamil Nadu Agricultural University, visited the Academy on February 27, 1999.
- A group of 134 final year B.Sc. (Agric.) and B.Sc. (Horticulture) students of Annamalai University, Annamalai Nagar, Tamil Nadu, visited the Academy on March 11, 1999.

Research / Review / Conference Papers

Chandel, B.S. and Chakaar, I.S. 1998. Economic evaluation of crop management research recommendation for farmers' adoption - A study on ginger cultivation in H.P. Indian J. Agric. Econ., 53, 53-60.

Chandel, B.S. and Jain, S.P. 1998. Do Panchayati Raj Institutions (PRIs) harbinger grassroot level growth in agricultural R & D process?: An enquiry through SWOT analysis, Indian J. Agric. Econ., 53, 465-477.

Kalpana Sastry, R. and Chattopadhyaya, C. 1999. Effect of soil solarization on *Fusarium oxysporum f. sp. carthami* populations in endemic soils. Indian Phytopathology, 52, 92-96.

Kalpana Sastry, R., Manikandan, P. and Anwer, M.M. 1999. Human resource management initiatives for excellence in government departments. Mgmt in Govt, 30, 59-72.

Katyal, J.C., Narayan Reddy, M. and Sharma, K.L. 1998. Current status of crop responses to fertilizers in rainfed areas - An experience from All India Co-ordinated Research Project for Dryland Agriculture, Fertilizer News, 44, 97-112.

Chattopadhyaya, C. and Kalpana Sastry, R. 1998. Retrospective and Perspectives of Oilseeds Pathology in India. In: Plant Pathology in India - 50 years (1947-1997). S. Chaudhari, C. Sen, and M.K. Dasgupta (eds.), Prof. S.B. Chattopadhyay festschrift vol., Dept Plant Pathology, Bidhan Chandra Krishi Viswavidyalaya, Kalyani, WB.

Chattopadhyaya, C. and Kalpana Sastry, R. 1999. Important Diseases of Sesame and Their Management Options. In: IPM Systems in Agriculture - 5: Oilseeds. R.K. Upadhyaya, K.G. Mukherji, and R.L. Rajak (eds.), Aditya Books Pvt. Ltd, Ansari Road, New Delhi, pp 387-398.

Kalpana Sastry, R. and Chattopadhyaya, C. 1999. Major Diseases of Safflower and Their Management. In: IPM Systems in Agriculture - 5: Oilseeds. R.K. Upadhyaya, K.G. Mukherji, and R.L. Rajak (eds.), Aditya Books Pvt. Ltd, Ansari Road, New Delhi, pp 234-256.

Lakshminarayana, M., Kalpana Sastry, R. and Prasad, M.V.R. 1999. Breeding for Pest Resistance in Oilseed Crops. In: IPM Systems in Agriculture - 5: Oilseeds. R.K. Upadhyaya, K.G. Mukherji, and R.L. Rajak (eds.), Aditya Books Pvt. Ltd., Ansari Road, New Delhi, pp 13-34.

Rao, K.P.C. and Evenson, R.E. 1998. Varietal Trait Values for Rice in India. In: Agricultural Values of Plant Genetic Resources. R.E. Evenson, D. Gollin, and V. Santaniello (eds.), CABI Publishing, Wallingford, U.K.

Chattopadhyaya, C. and Kalpana Sastry, R. 1999. Epidemiology of alternaria blight of safflower and sunflower. Paper presented in Annual Conference of Indian Society of Mycology and Plant Pathology, held at AAU, Jorhat, March 26-27, 1999.

Chattopadhyaya, C. and Kalpana Sastry, R. 1999. Relationship among pathogen population in soil, disease incidence, plant height, and yield for two diseases of sesame and safflower. Paper presented in Annual Conference of Indian Society of Mycology and Plant Pathology held at AAU, Jorhat, March 26-27, 1999.

Kalpana Sastry, R., Chattopadhyaya, C., Vrijendra Singh, and Hedge D.M. 1998. Combined host resistance, cultural and chemical measures to manage safflower wilt. Presented in Symposium on Microbial Diversity and Host Pathogen Interactions, Organized by Indian Phytopathological Society and UGC-SAP, held at Dept of Botany, Osmania University, Dec 19, 1998.

Kalpana Sastry, R. and Chattopadhyaya, C. 1999. Development of wilt resistance lines in safflower. Paper presented in Annual Conference of Indian Society of Mycology and Plant Pathology, held at AAU, Jorhat, March 26-27, 1999.

Katyal, J.C. 1998. Scientists' perception for agriculture 2020 on soil, water, and environmental sciences. Paper presented at the Brainstorming Session on Scientists' Perception for Agriculture - 2020, held at NBPGR, New Delhi, June 4 - 5, 1998.

Katyal, J.C. and Balaguru, T. 1998. Education - research - extension linkages. Paper presented in the Symposium on Agronomy, Environment, and Food Security for 21st Century, held at New Delhi, November 23-27, 1998.

Katyal, J.C. and Jagannadham Challa. 1998. Organizational structure and autonomy in ICAR, ICMR, and CSIR - a brief comparative analysis. Paper presented at the Brainstorming Session on O & M Task Force of NATP, held at ICAR, New Delhi, November 26 and 27, 1998.

Rajeswari, B., Chandrasekhara Rao, K., Pramod Kumar, C., Shantaram, M.M. and Kalpana Sastry, R. 1997. Integrated Disease Management (IDM) for dry root rot of mungbean under field conditions. Abstract No. p8-020, Indian Phytopathological Society Golden Jubilee International Conference, held at New Delhi, November 10 -15, 1997.

Reports and Manuals

Katyal, J.C., Rao, K.P.C. and Manikandan, P. 1998. Project Based Budgeting, RPF, and AAI - Guidelines and Proformae. NAARM, Rajendranagar, Hyderabad.

Rama Rao, D. and Nanda, S.K. 1998. Training Manual on Technological Forecasting and Assessment. NAARM, Hyderabad.

Karna Rao, D. and Sandhya Shenoy, N. 1998. Recommendations of the Workshop on Home Science Education. NAARM, Hyderabad.

Karna Rao, D. and Sandhya Shenoy, N. 1998. Professional Women in Agriculture. DST Project Report. NAARM, Hyderabad.

Karna Rao, D. and Vidyasagar Rao, K. 1998. Software Manual on Forecasting Packages. NAARM, Hyderabad.

Books

Prasad, C. and Balaguru, T. (eds) 1999. Women in Agriculture - Developmental Issues. NCW, IAWA, and NAARM, Hyderabad, 388 pp.

Resource Papers Developed

- Production Systems Approach - PSA (T. Balaguru)
- Implementation of PSA for Sustainable Agricultural Development - Issues (T. Balaguru and P. Manikandan)
- Status Descriptors of Teachers in Agricultural Universities (A. Gopalam and R. Kalpana Sastry)
- Distance Training for Agricultural Research Management (Jagannadham Challa and Nicole Brunet)
- Human Resource Development for Agricultural Education (Jagannadham Challa)
- Ecosustainability, Natural and Human Resources (J.C. Katyal)
- Institute Management (J.C. Katyal and T. Balaguru)
- Adult Learning Strategies (P. Kumar)
- Dynamics of Labour Management (R. Kalpana Sastry)
- Basics in Powerpoint (R. Kalpana Sastry)
- Formative Evaluation Techniques (R. Kalpana Sastry)
- Management of Change (P. Manikandan)
- Conflicts and Negotiation (P. Manikandan)
- Management of Agricultural Education (S.N. Saha)
- Creative Techniques for Problem Solving (S.N. Saha)
- Leadership Development in Agricultural Education (S.N. Saha)

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Obituary

We record, with profound grief, the demise of Dr K. Venkateswarlu, Principal Scientist, who passed away on November 8, 1998. He is survived by his wife and one son.

Dr K. Venkateswarlu had his B.Sc (Agric.) from Andhra Pradesh Agricultural University and Masters Degree from Agra University. He obtained his Ph.D. in Agronomy from G.B. Pant University of Agriculture and Technology, Pantnagar. Dr Venkateswarlu joined the Academy as a faculty member on May 28, 1979. Earlier to joining the Academy, he worked as Scientist S-1 at CSWRI, Avikanagar. During his illustrious career of about two decades at the Academy, Dr Venkateswarlu had advanced training in management from IIM, Ahmedabad, and also from USA. Dr Venkateswarlu was one of the effective teaching faculty members of the Academy. Apart from teaching, Dr Venkateswarlu was also actively involved in research and consultancy. His major areas of interest in the Academy included Project management techniques, Management of research stations, and SWOT analysis.



Dr Venkateswarlu's death came as a terrible shock to the members of the NAARM family and it resulted in a great loss to the Academy. Dr Venkateswarlu's memory will always be treasured by all those who had the privilege of knowing his sunny disposition, his fine mind, and his affectionate heart.



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